

Associativity of Multiplication is Hard for Resolution

Vincent Liew

Circuits for $x + y$

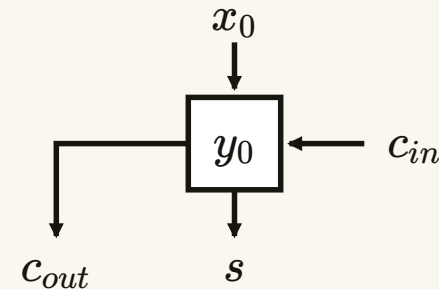
Length 1:

- Use **full adder** circuit

$$c_{out} = \text{MAJ}(x_0, y_0, c_{in})$$

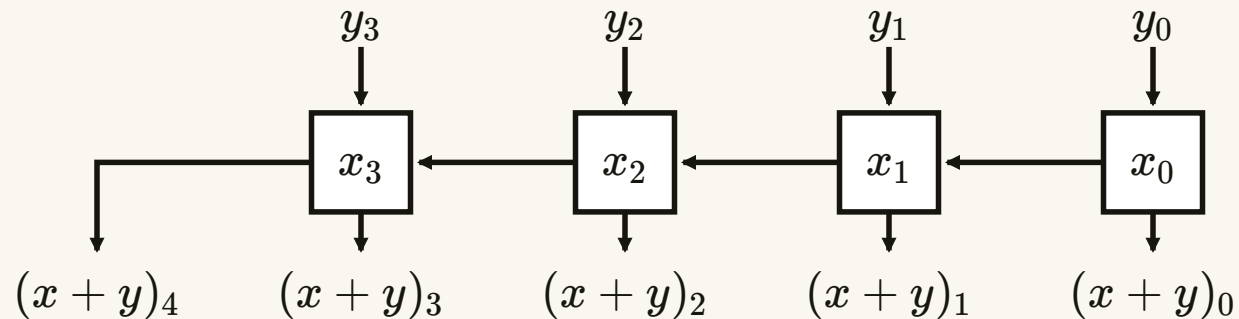
$$s = x_0 \oplus y_0 \oplus c_{in}$$

- Outputs sum in binary: $2c_{out} + s = x_0 + y_0 + c_{in}$



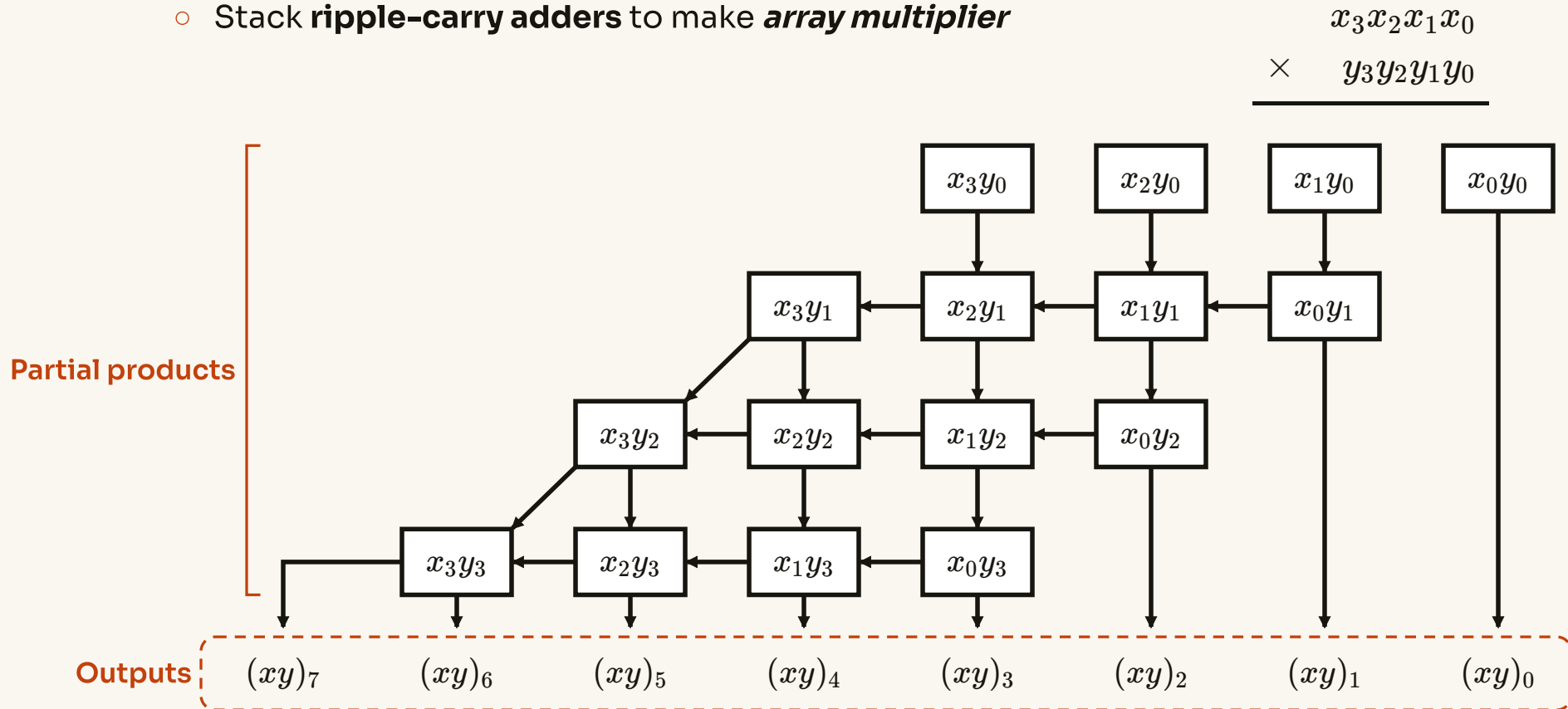
Length n :

- Chain **full adders** to form **ripple-carry adder** circuit

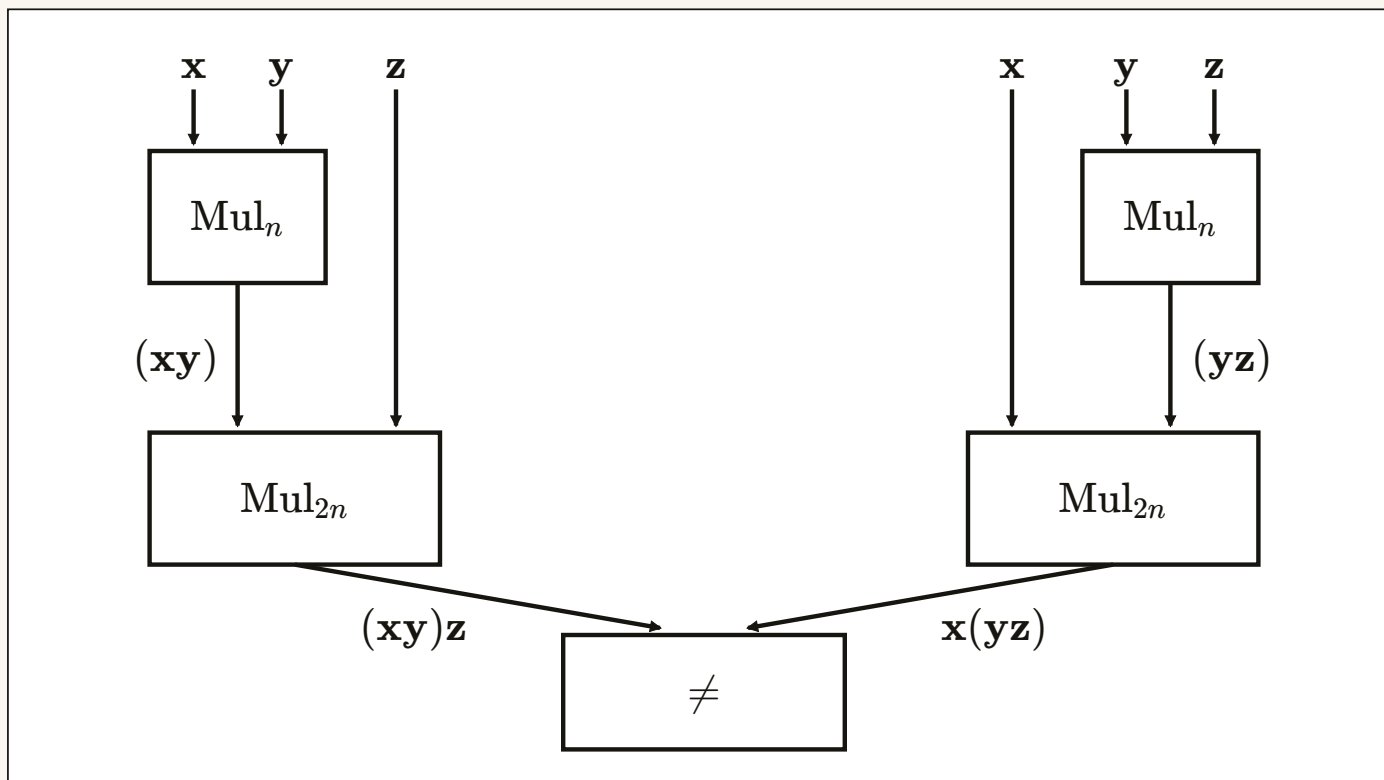


Circuits for $x \times y$

- Stack **ripple-carry adders** to make **array multiplier**



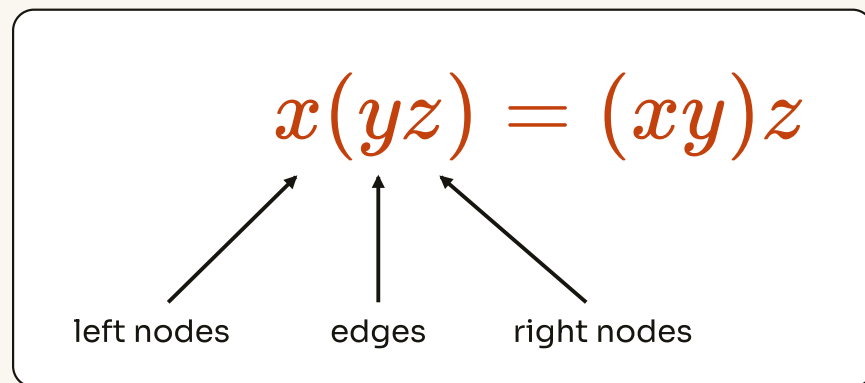
The miter Assoc_n



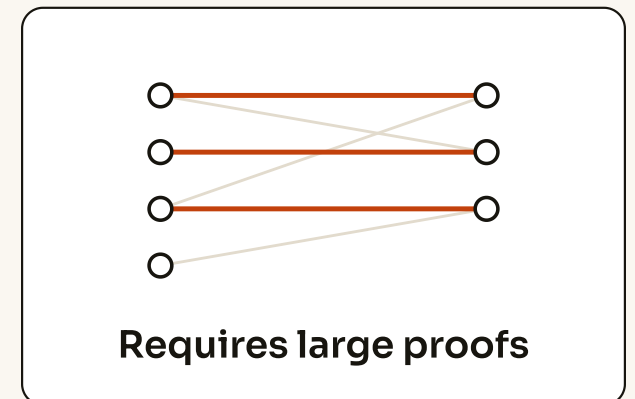
Main theorem

Theorem. $\text{ResSize}(\text{Assoc}_n) \geq 2^{\Omega((n/\log n)^{1/4})}$

Key insight: embed perfect matching contradiction inside Assoc_n

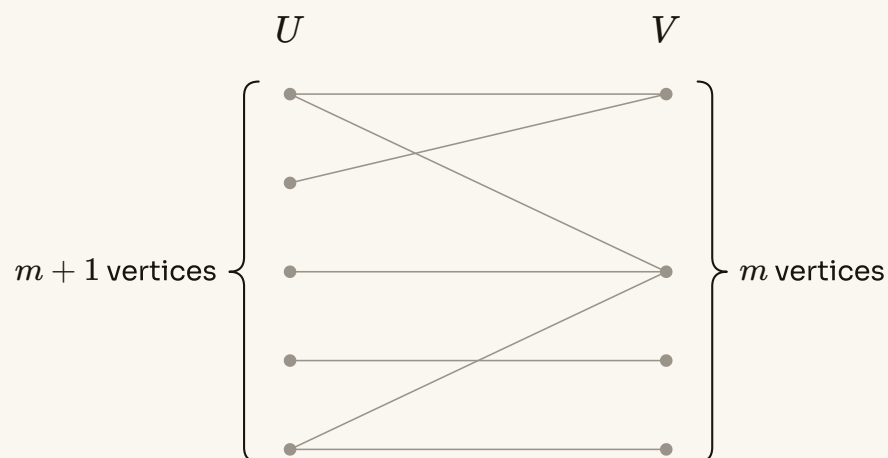



Set bits of x, y, z



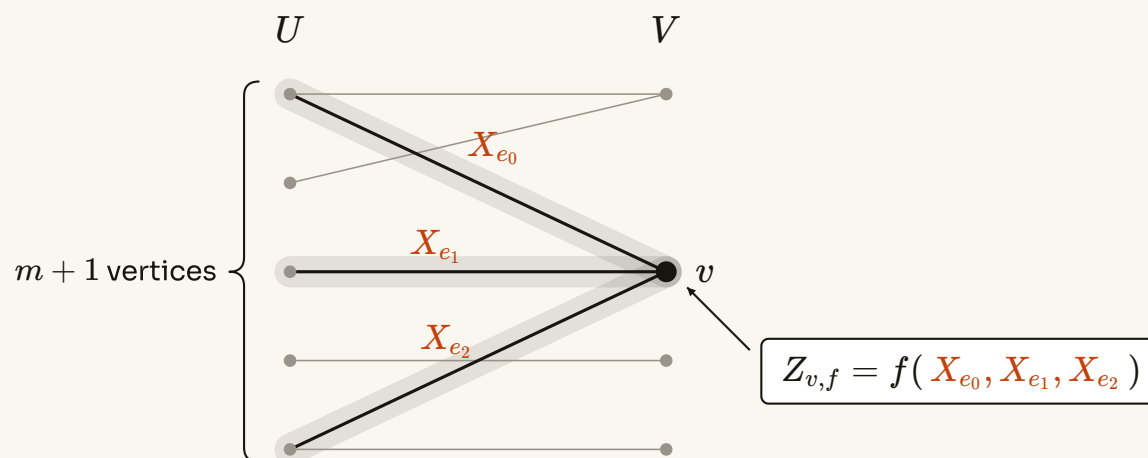
Star-local perfect matching (SPM)

- Want bipartite graph of bounded degree where perfect matching is hard
 - Expanders from [ISS2015] require $2^{\Omega(m)}$ size proofs, proven via width



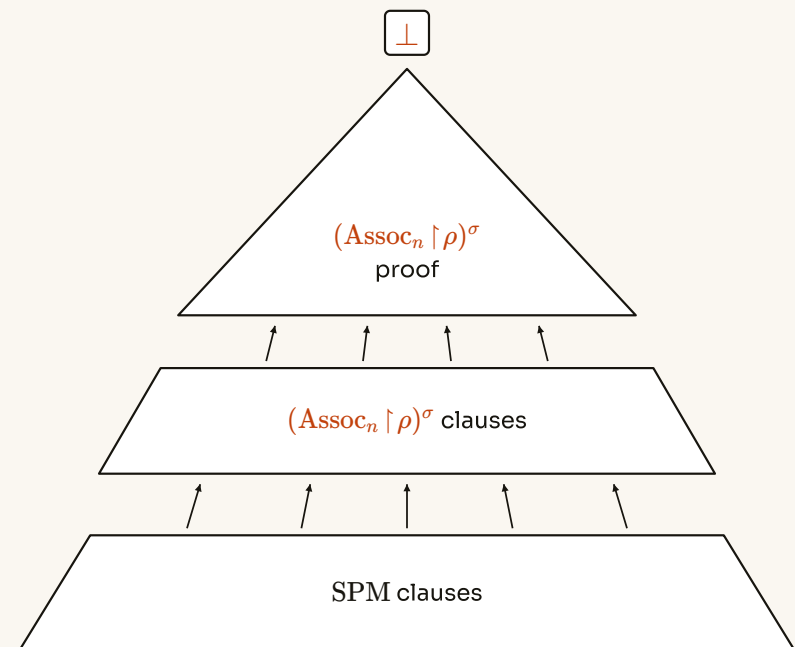
Star-local perfect matching (SPM)

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 - Expanders from [ISS2015] require $2^{\Omega(m)}$ size proofs, proven via width
 - Bounded degree $\rightarrow$ constant factor increase in # variables/clauses
 - Size-width tradeoff still gives exponential $2^{\Omega(m)}$ lower bound



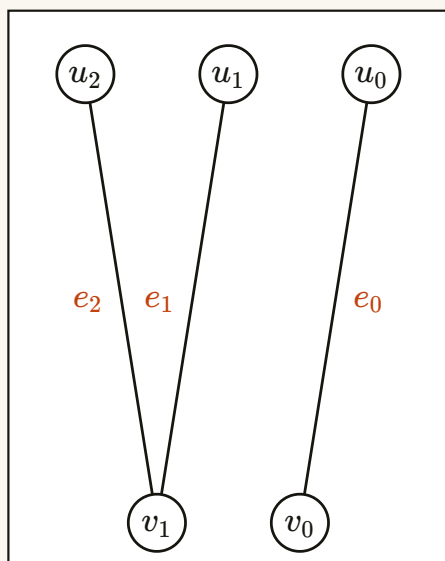
Bird's eye view of reduction

- **(restrict)** Construct restriction ρ so $\text{Assoc}_n \upharpoonright \rho$ enforces perfect matching constraints on subset of $\mathbf{y}$ variables
- **(substitute)** Define σ replacing circuit variables with SPM extension variables
- **(compose)** Use SPM clauses to derive each clause of $(\text{Assoc}_n \upharpoonright \rho)^\sigma$



g -sparse restriction of bit-vectors for $g = 4$

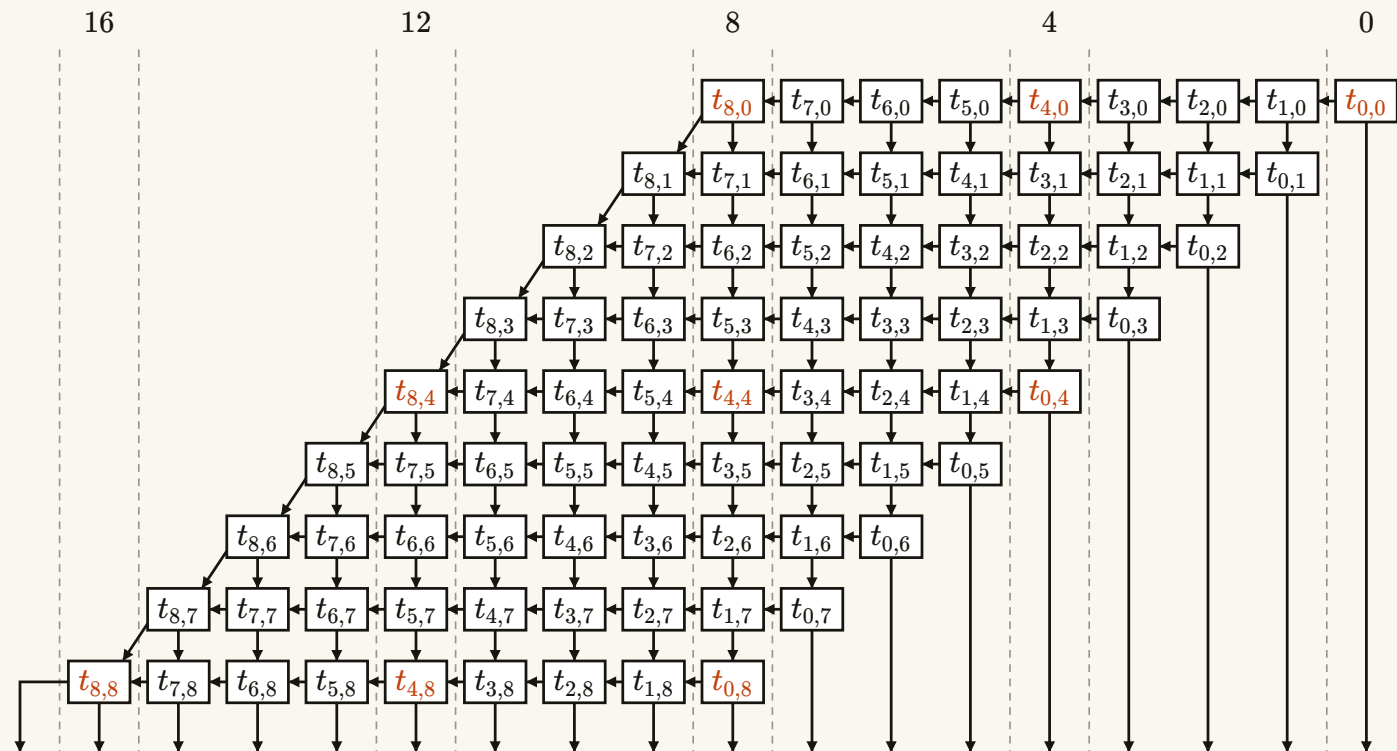
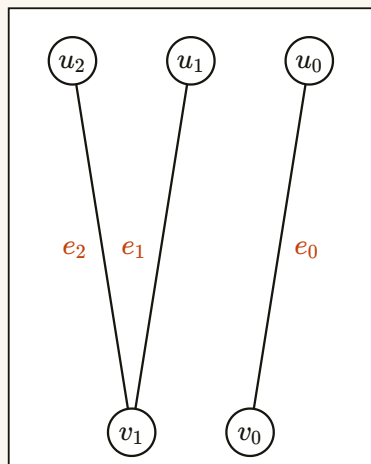
Restriction ρ is g -sparse if it sets all non-multiple-of- g positions to 0



	8				4				0
x:	1	0	0	0	1	0	0	0	1
	↑ u_2				↑ u_1				↑ u_0
y:	X_{e_2}	0	0	0	X_{e_1}	0	0	0	X_{e_0}
	↑ e_2				↑ e_1				↑ e_0
z:	1	0	0	0	0	0	0	0	1
	↑ v_1								↑ v_0

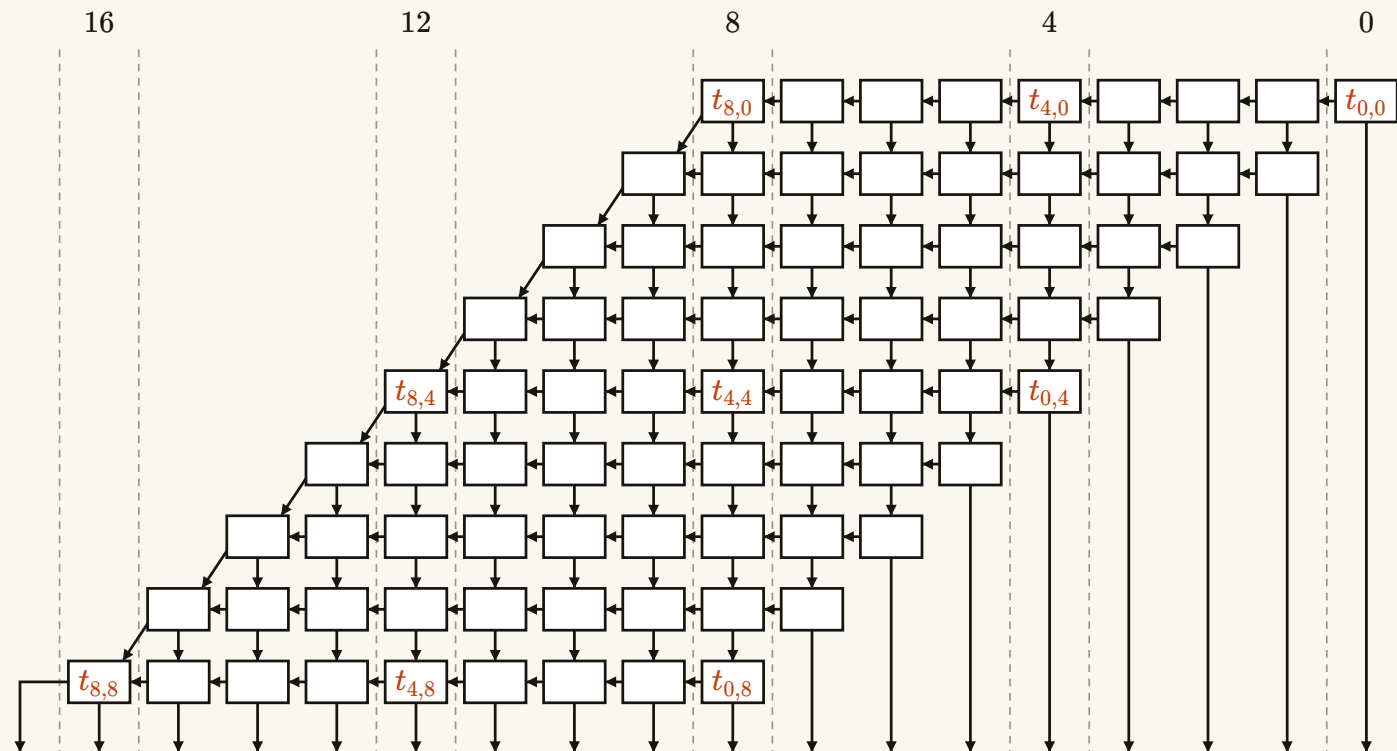
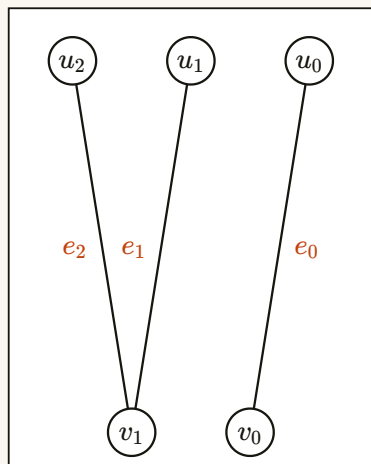
g -sparse restrictions for $g = 4$

$\mathbf{x}$:	1	0	0	0	1	0	0	0	1
$\mathbf{y}$:	X_{e_2}	0	0	0	X_{e_1}	0	0	0	X_{e_0}
$\mathbf{z}$:	1	0	0	0	0	0	0	0	1



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x:

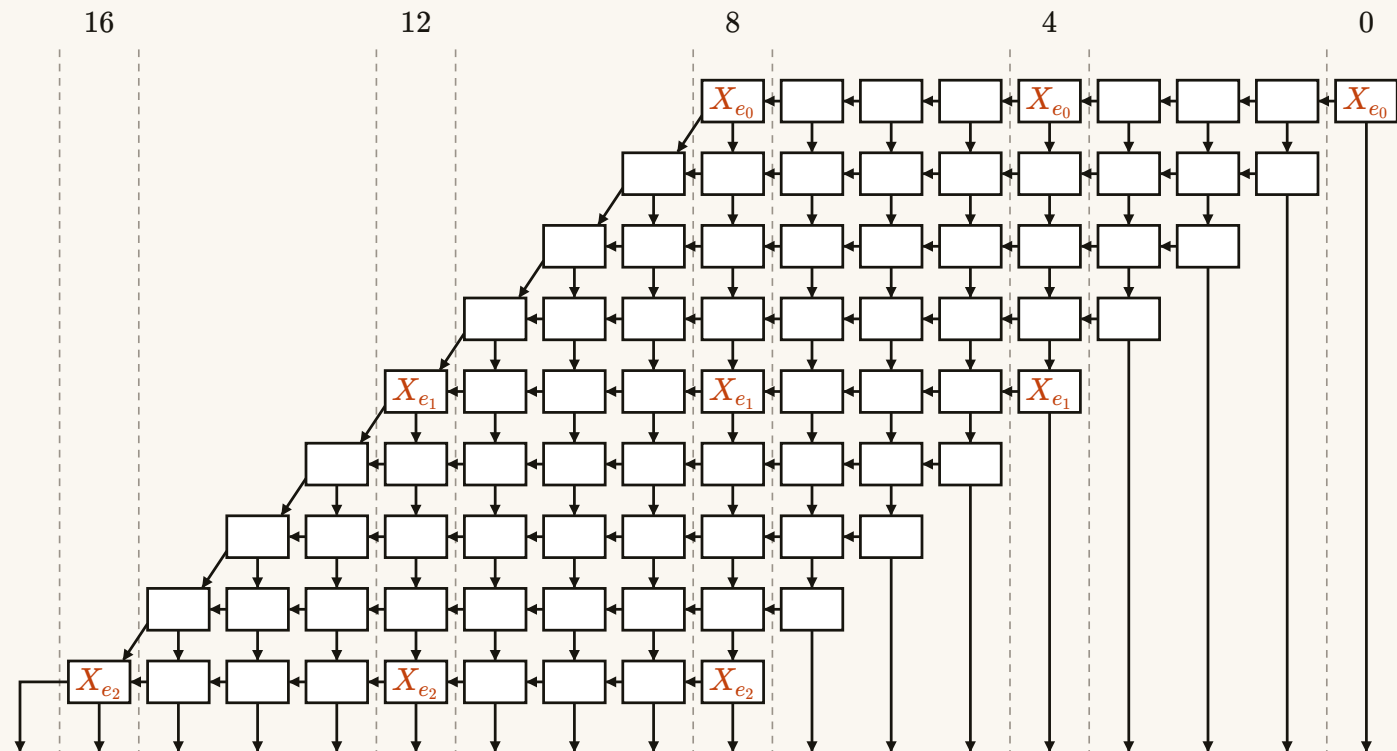
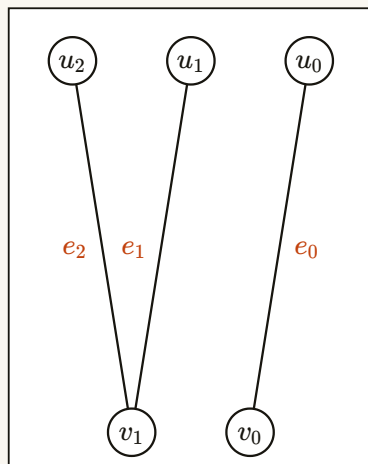
1	0	0	0	1	0	0	0	1
---	---	---	---	---	---	---	---	---

y:

X_{e_2}	0	0	0	X_{e_1}	0	0	0	X_{e_0}
-----------	---	---	---	-----------	---	---	---	-----------

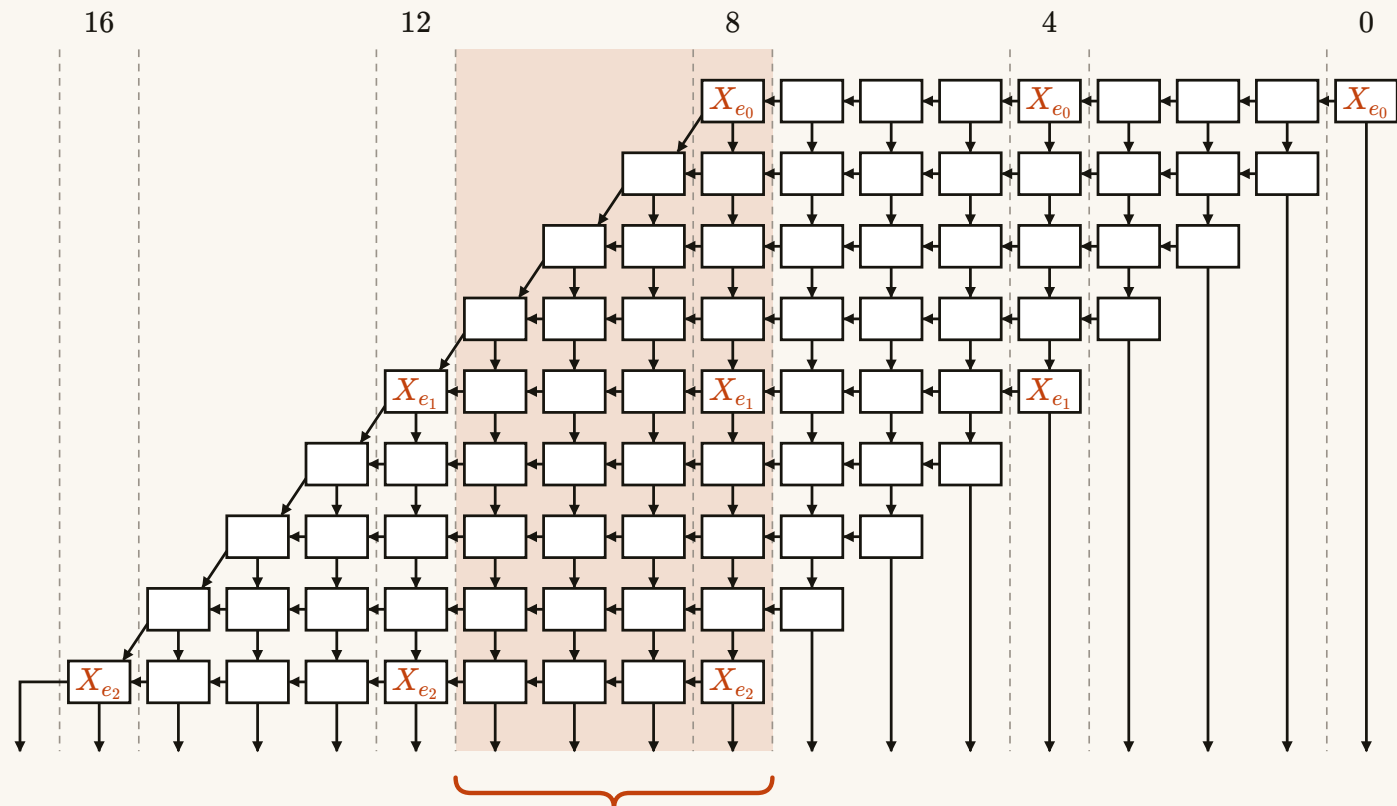
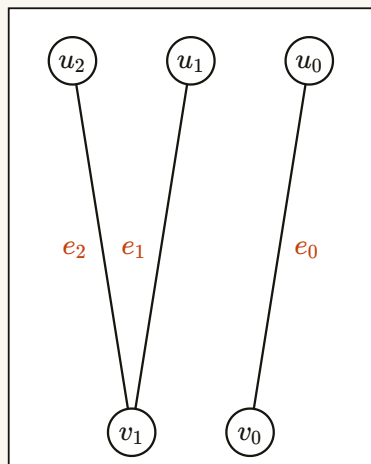
z:

1	0	0	0	0	0	0	0	1
---	---	---	---	---	---	---	---	---



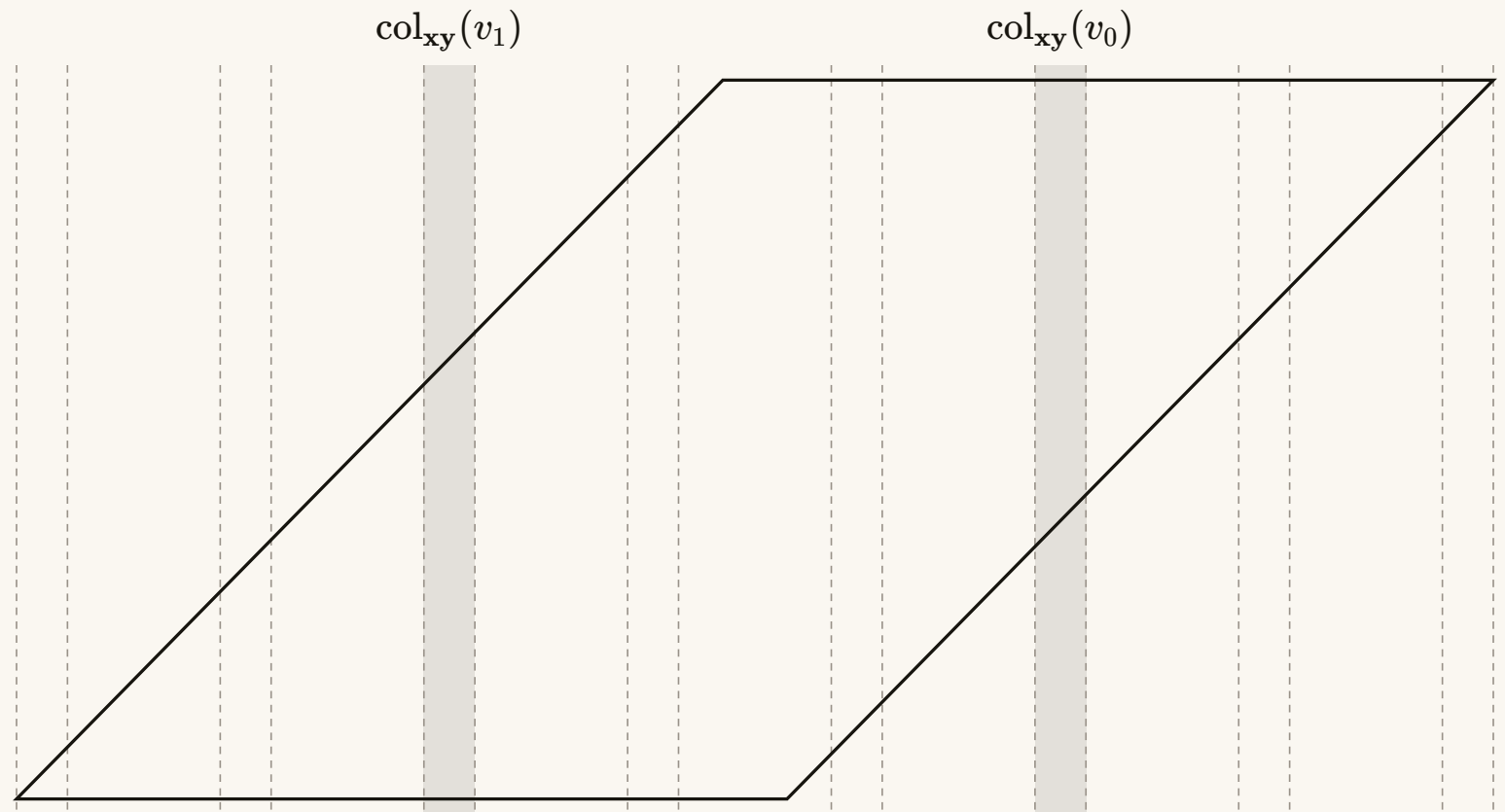
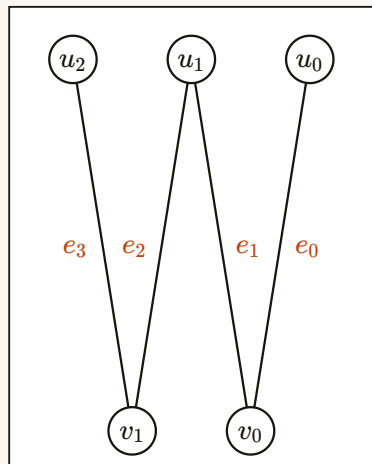
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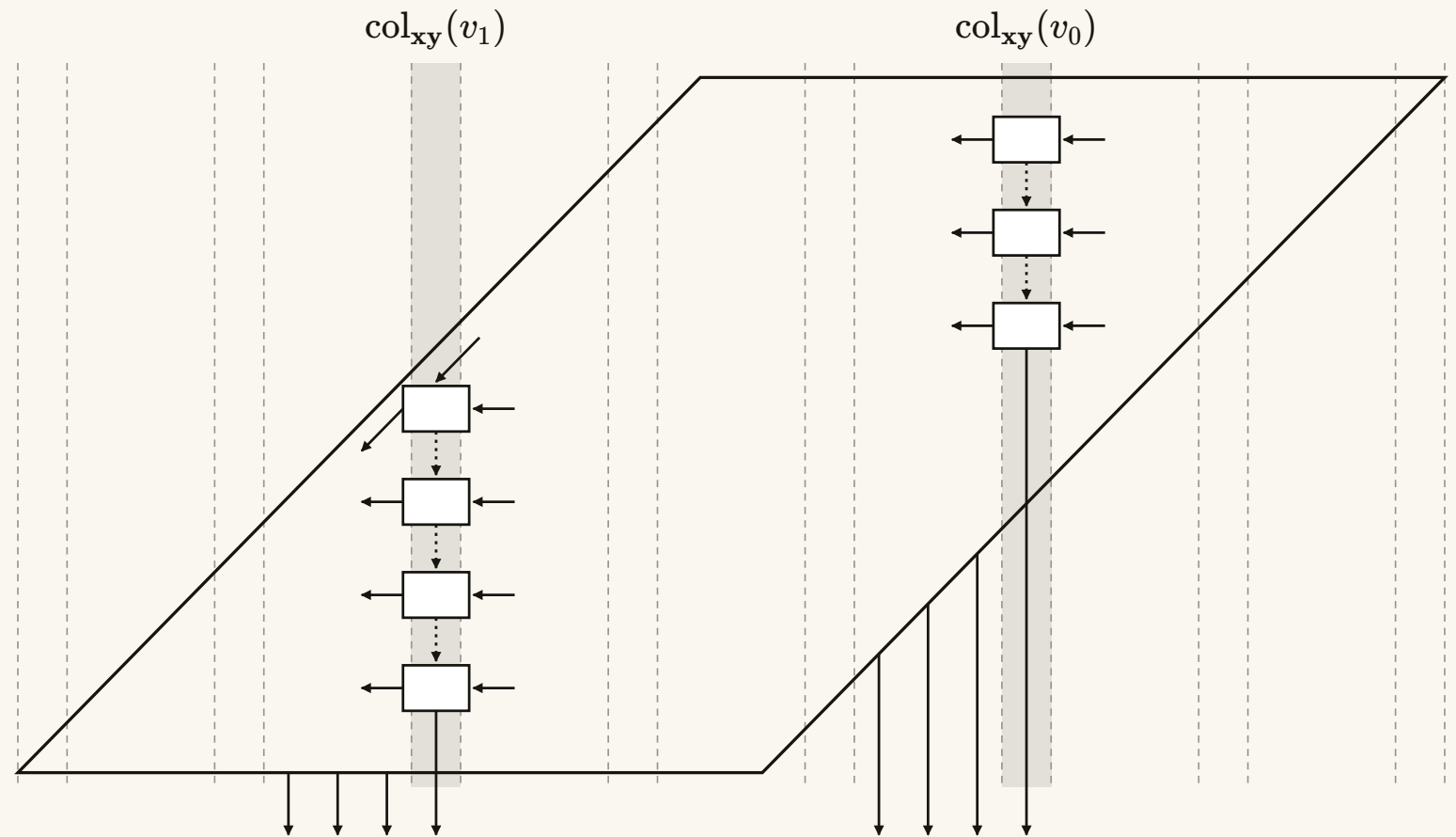
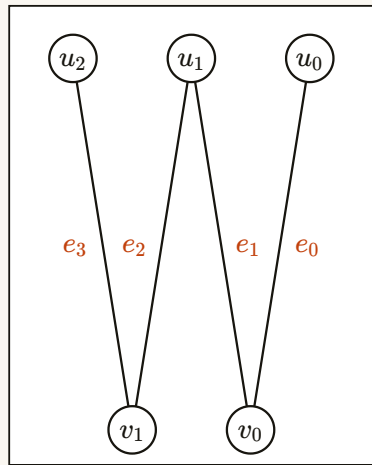


- outputs $\sum_e X_e$ in binary
- wires only depend on $X_{e_0}, X_{e_1}, X_{e_2}$

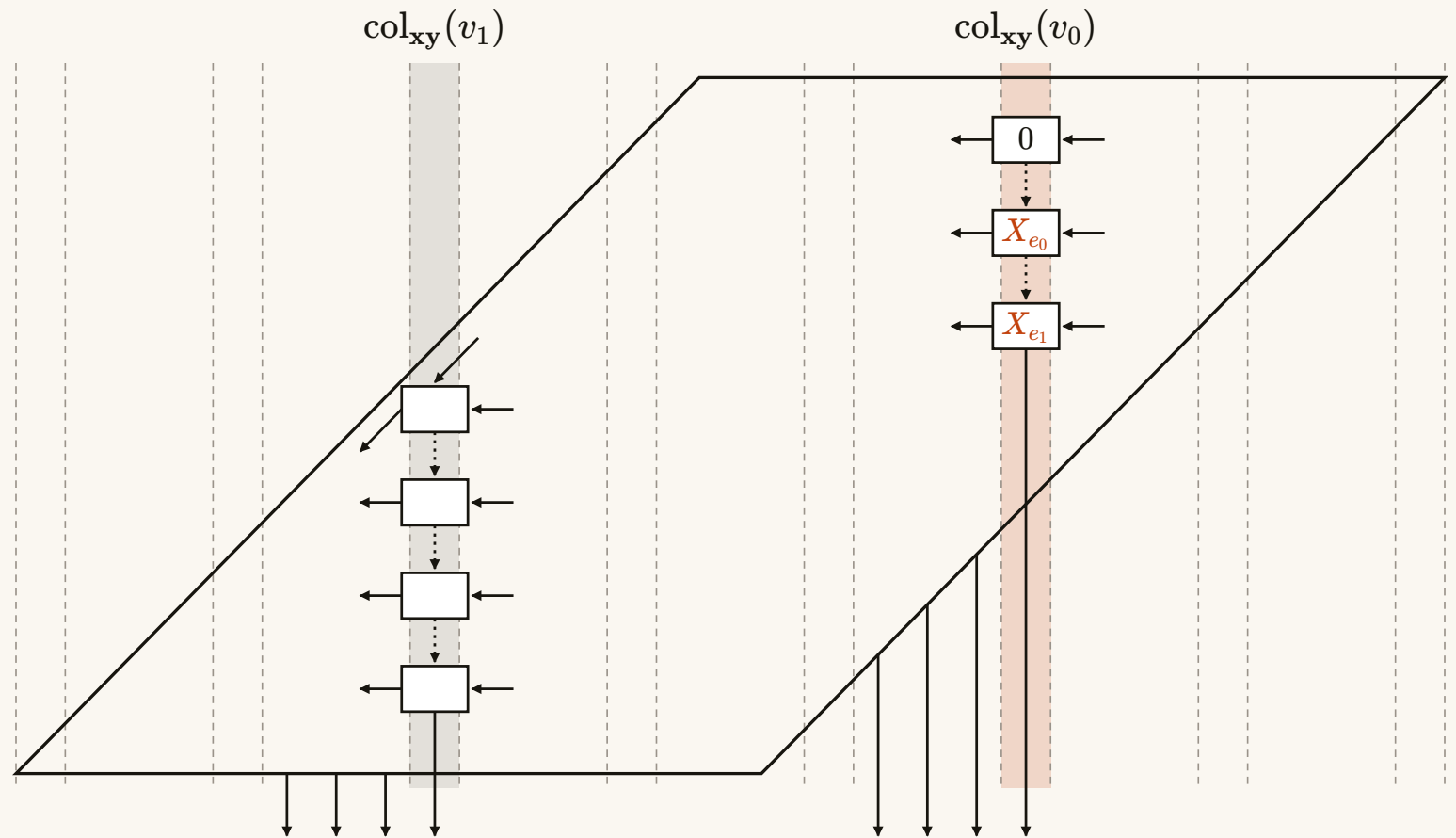
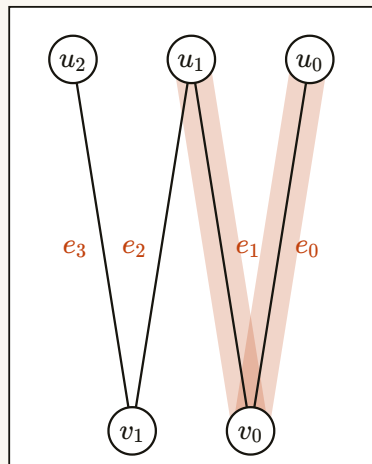
Restriction ρ in xy



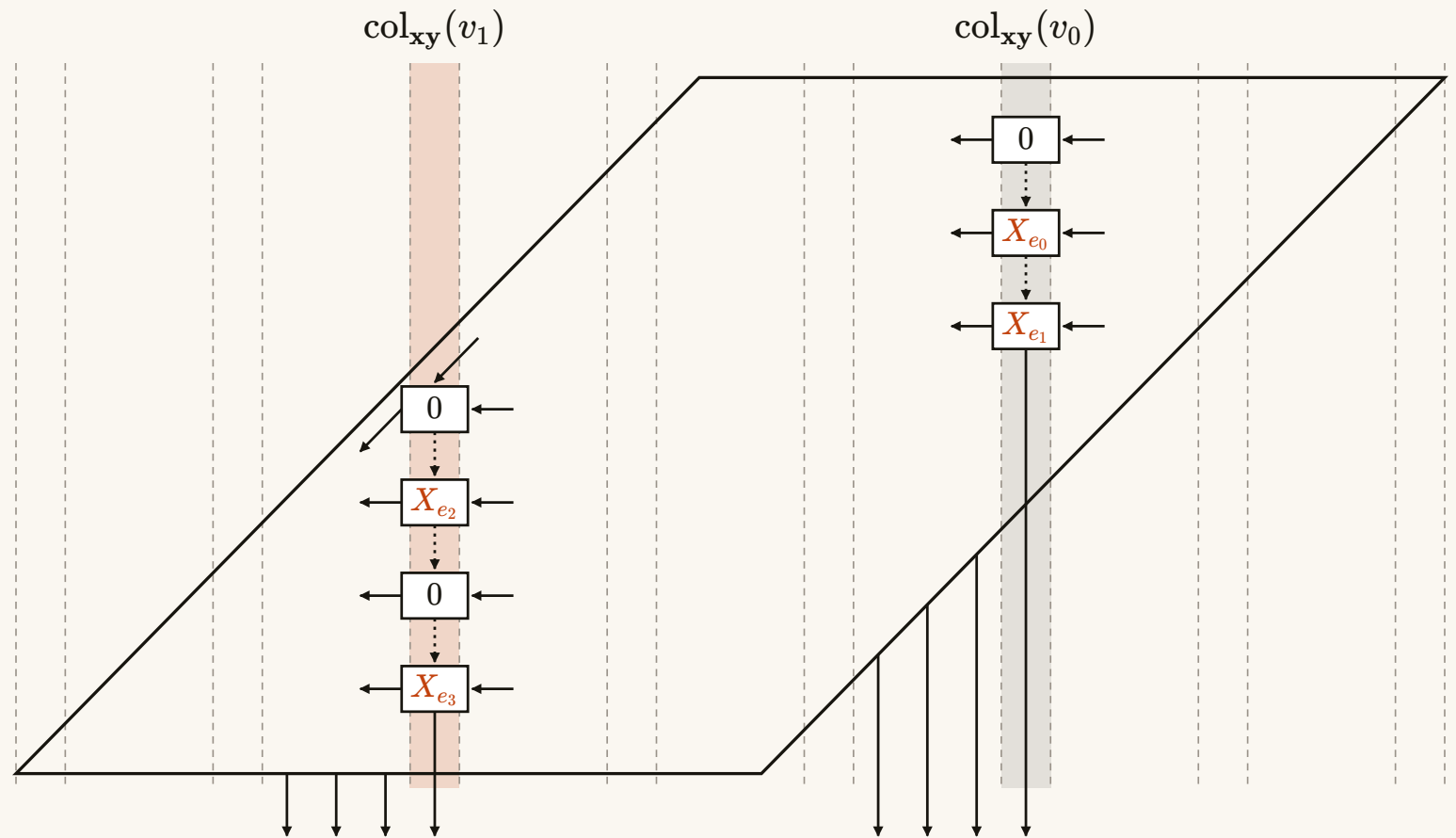
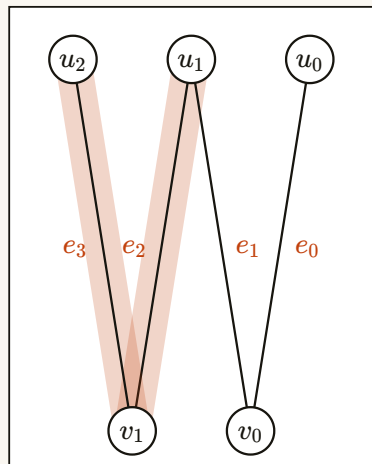
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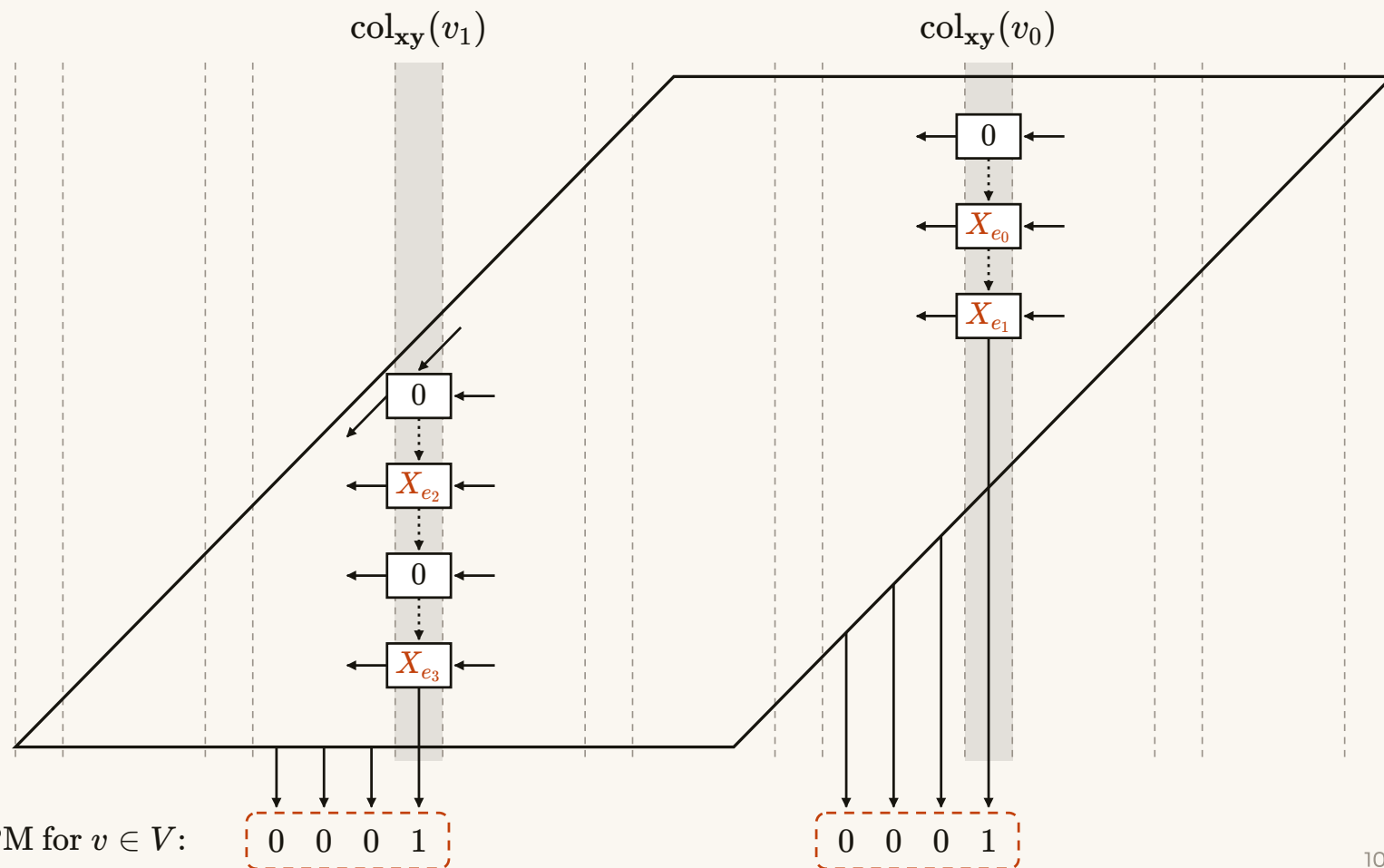
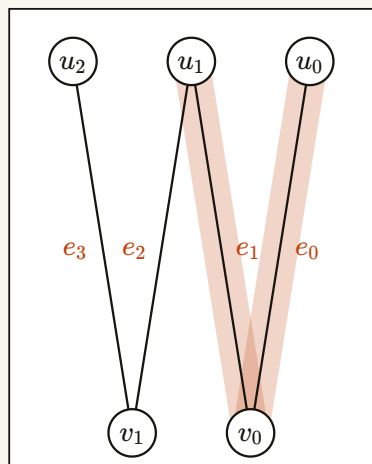
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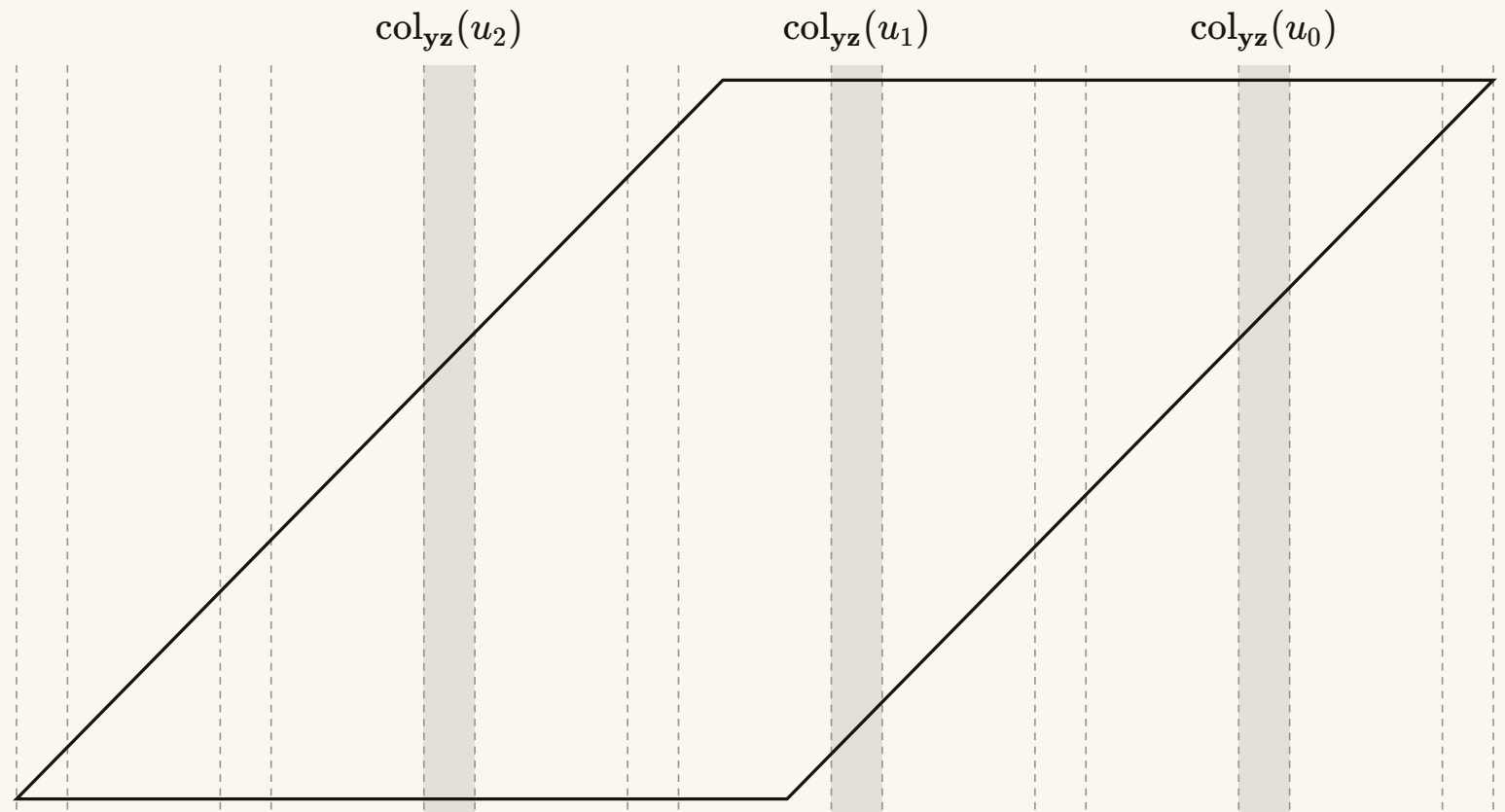
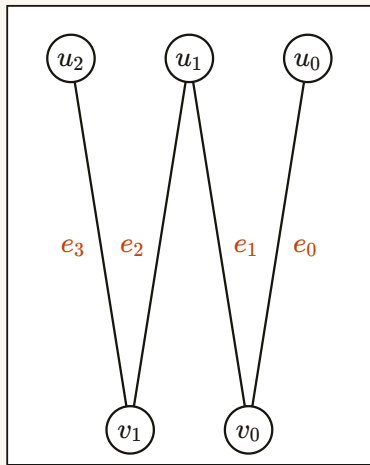
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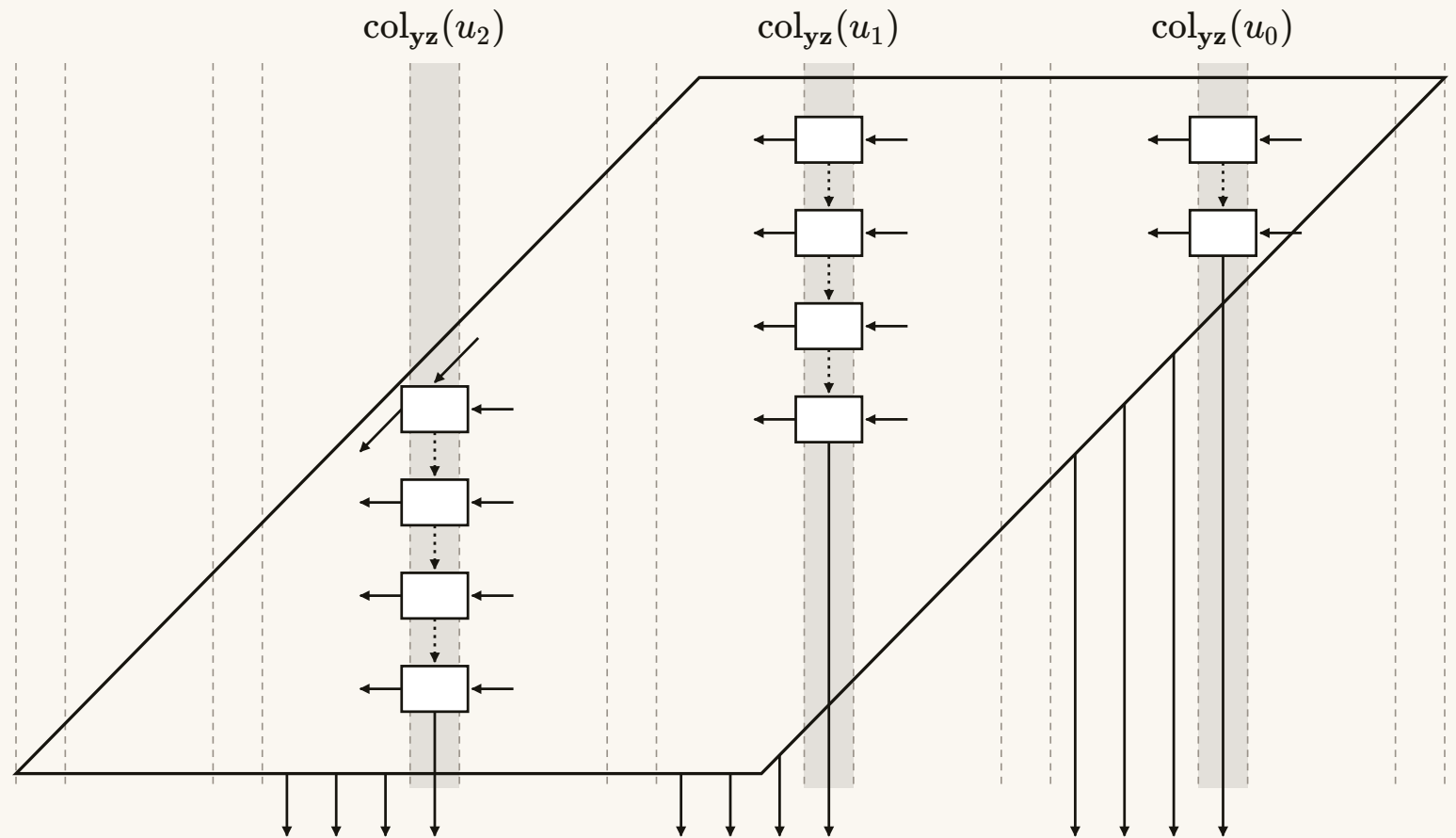
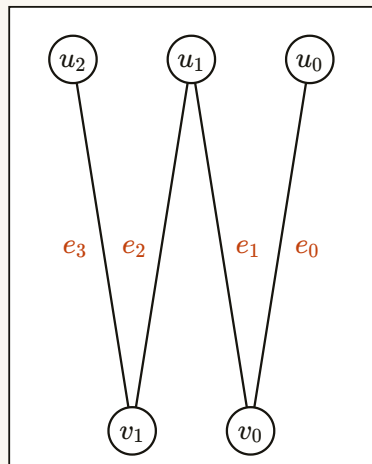
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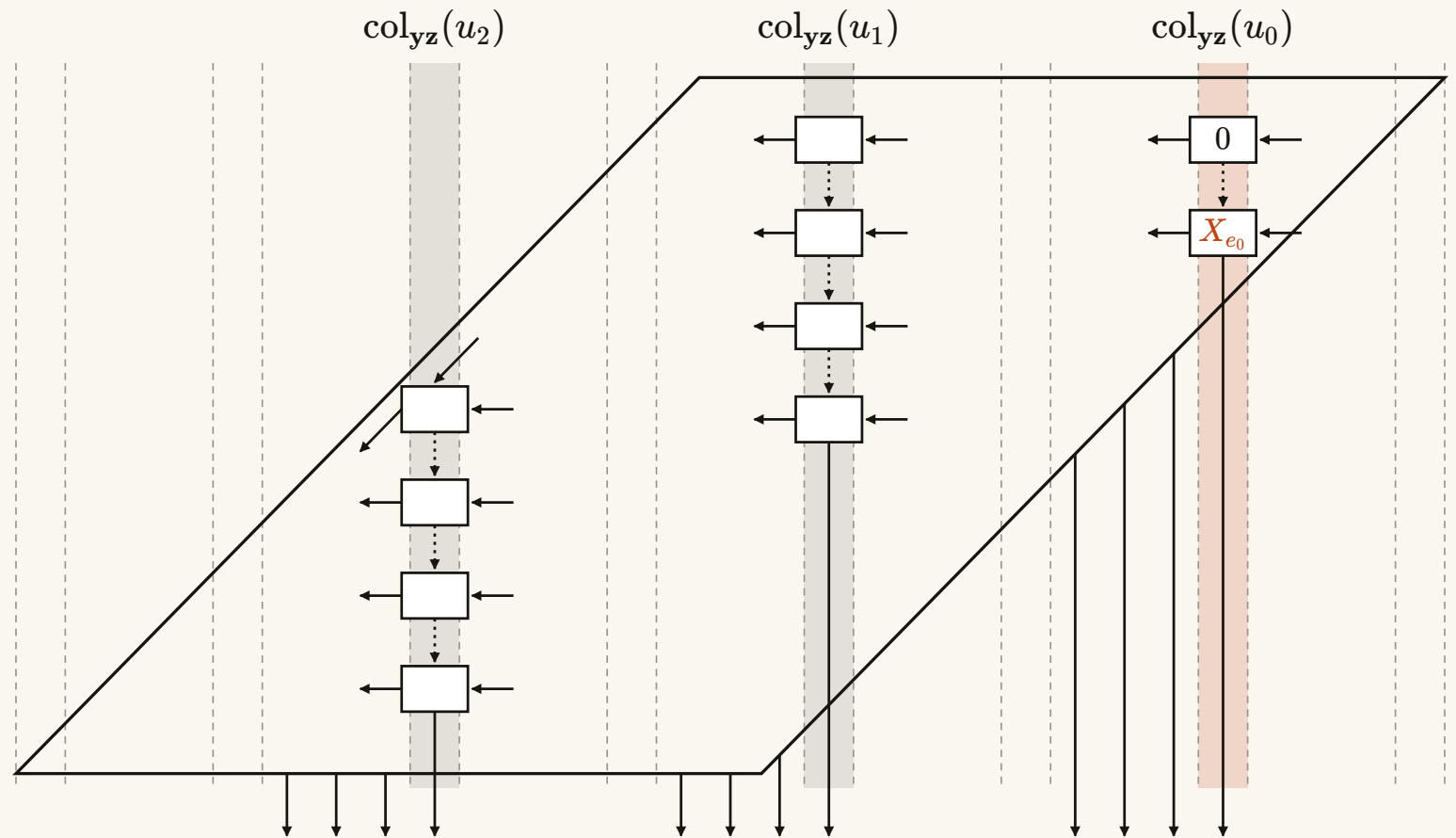
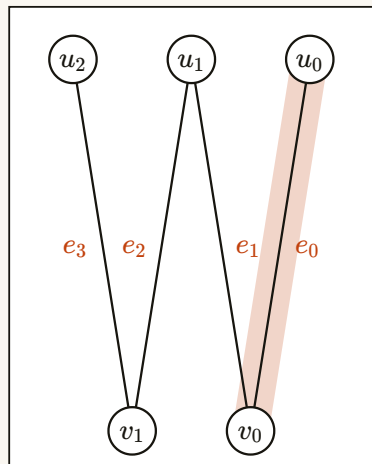
Restriction ρ in yz



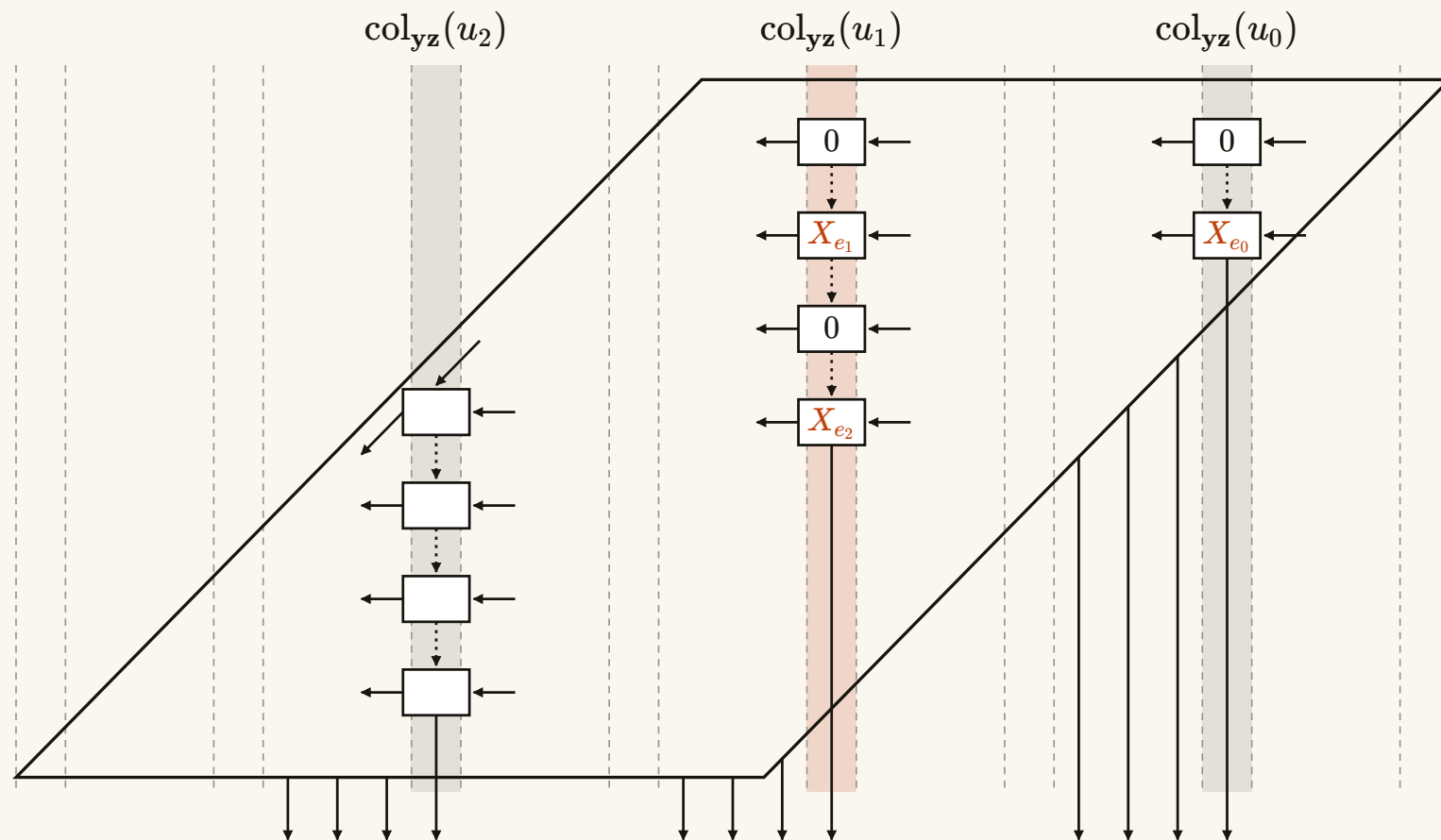
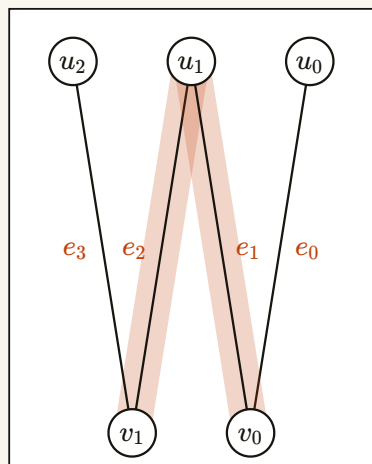
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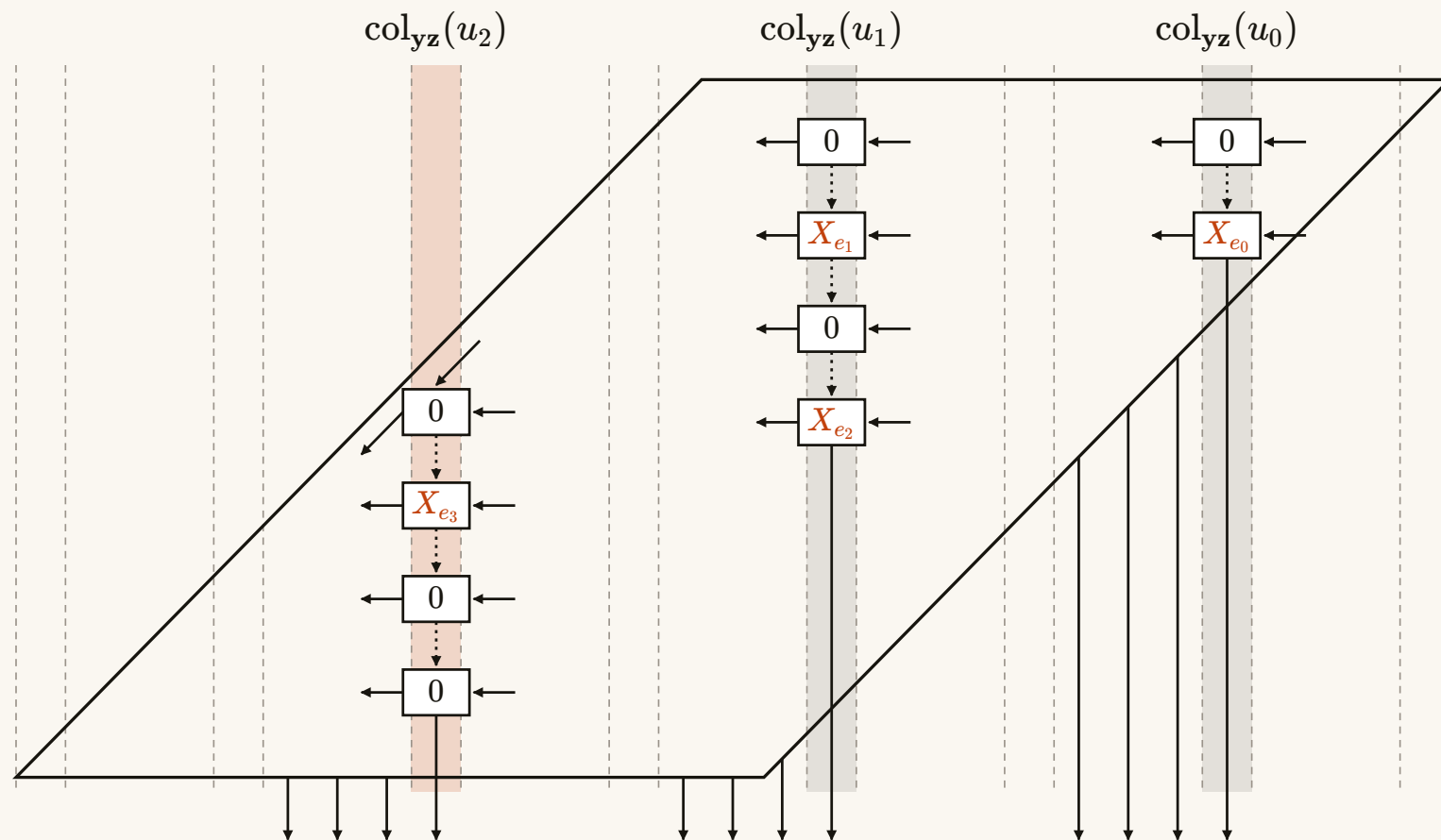
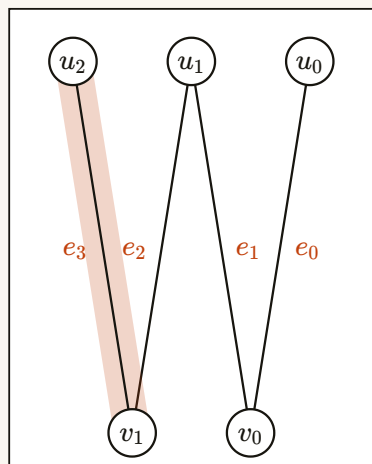
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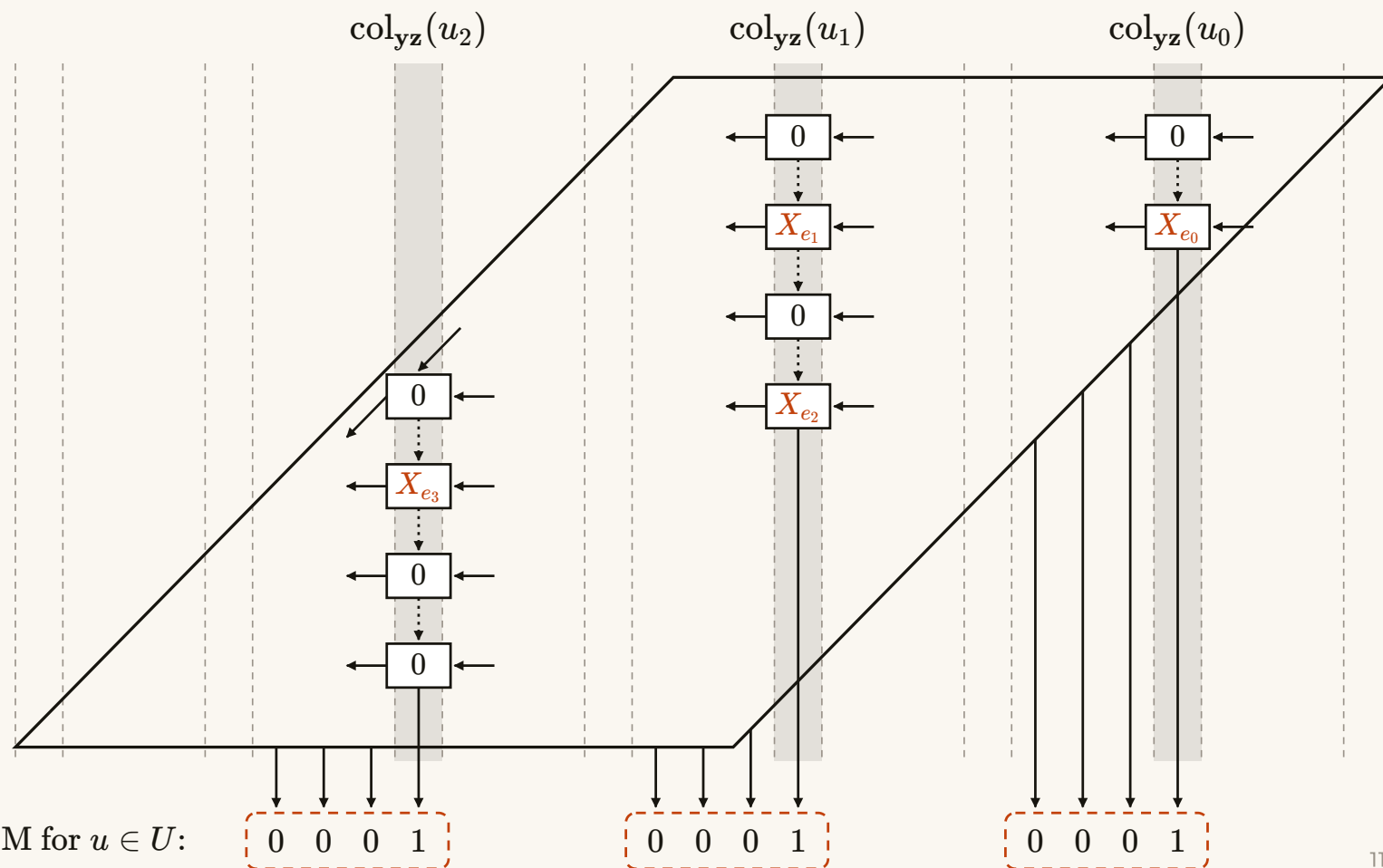
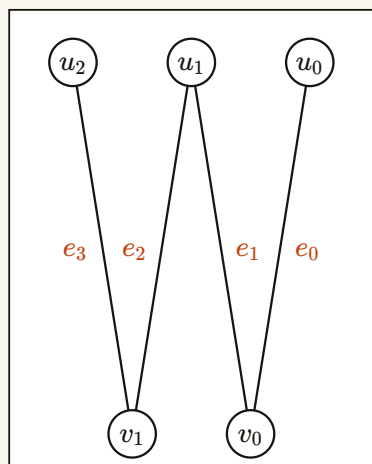
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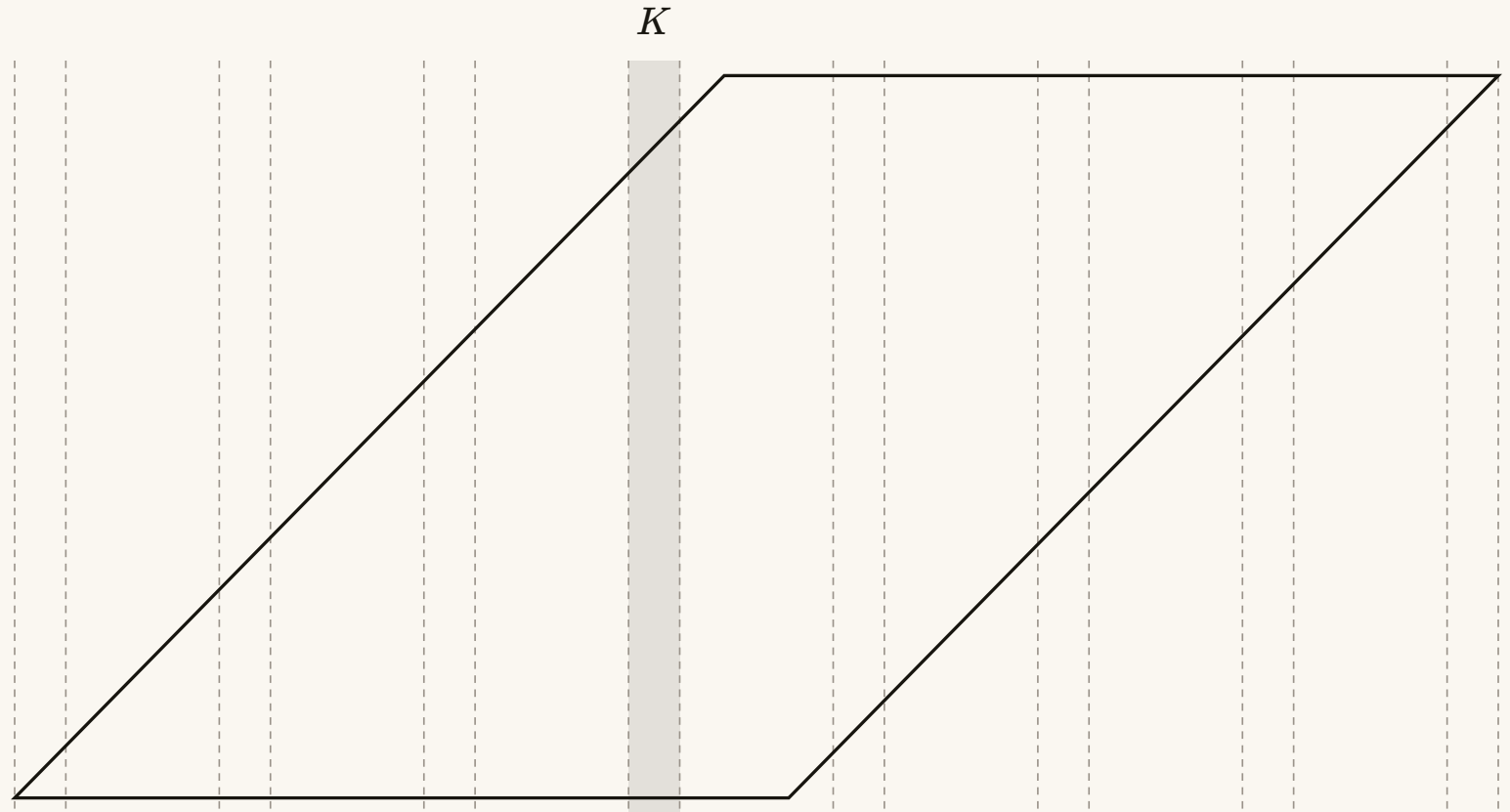
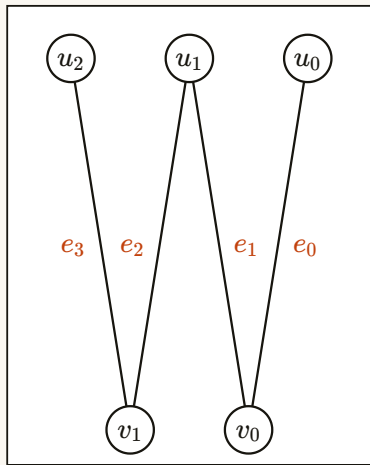
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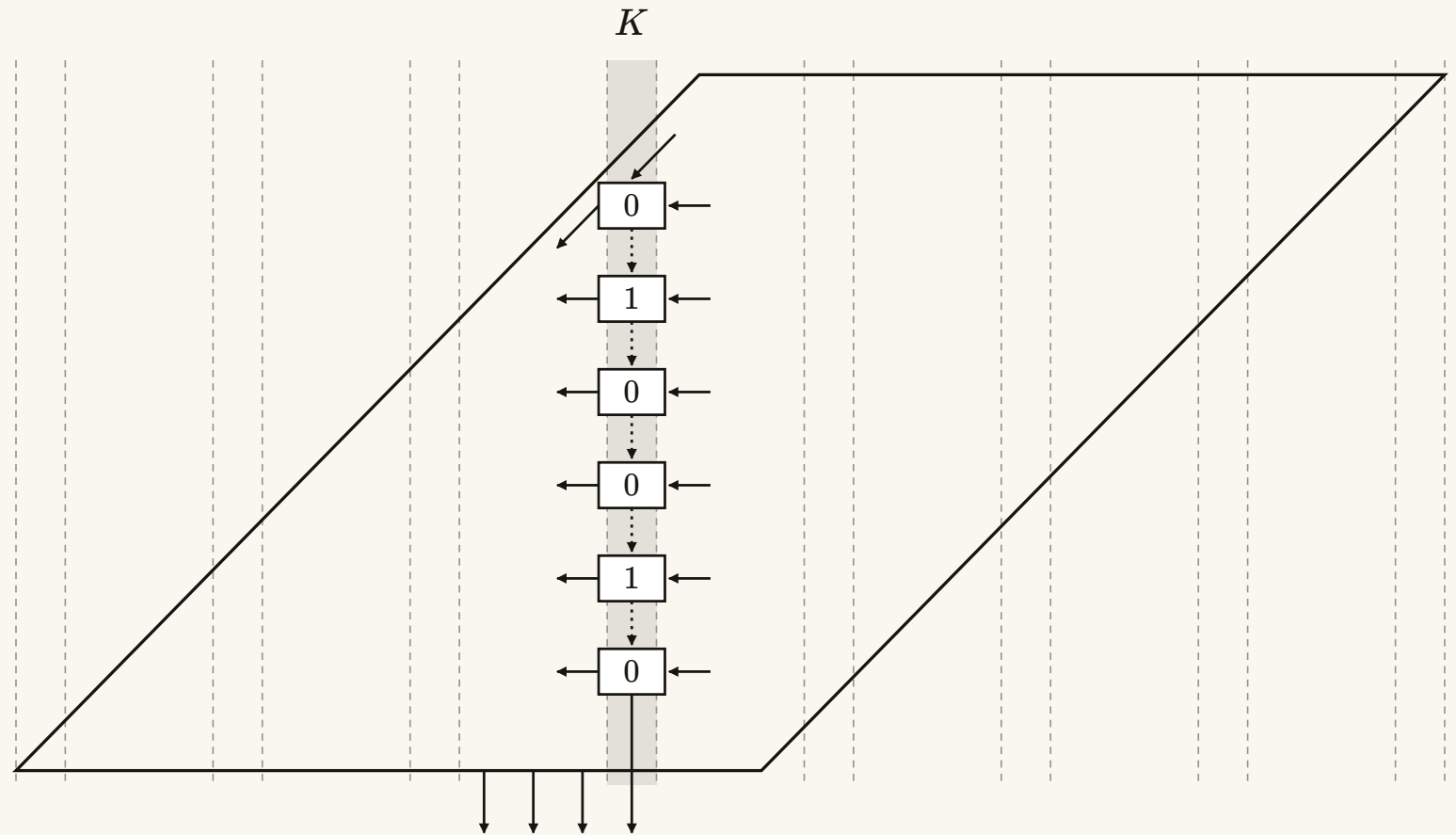
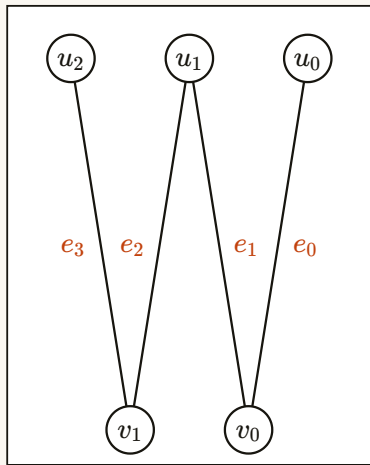
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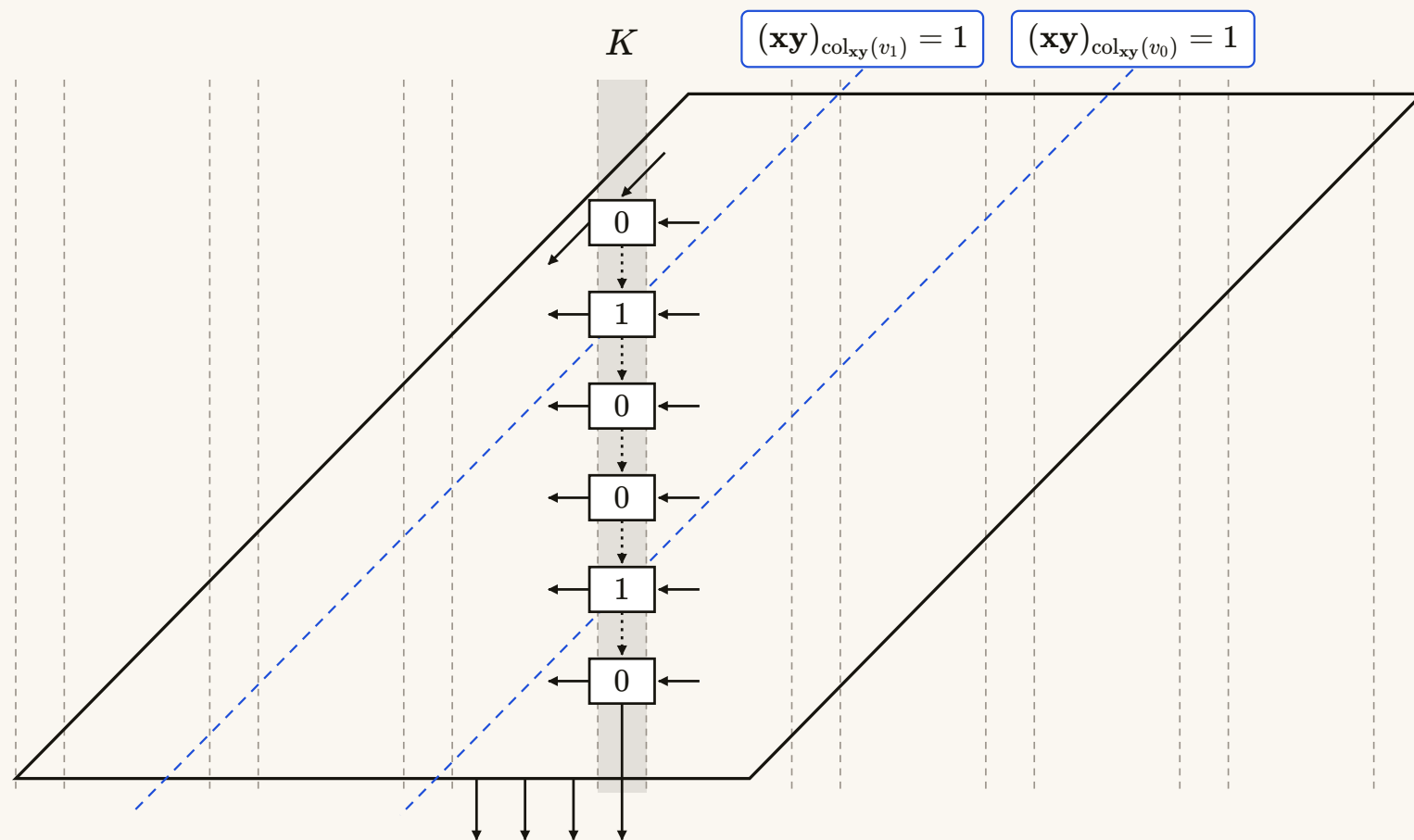
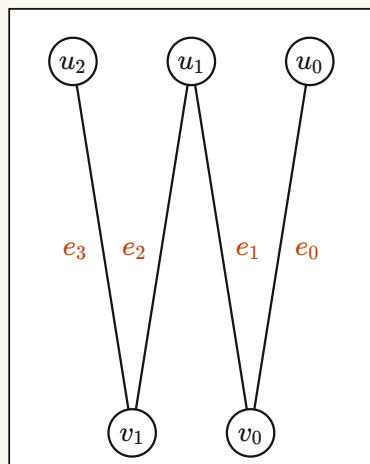
Restriction ρ in $(\mathbf{xy})\mathbf{z}$



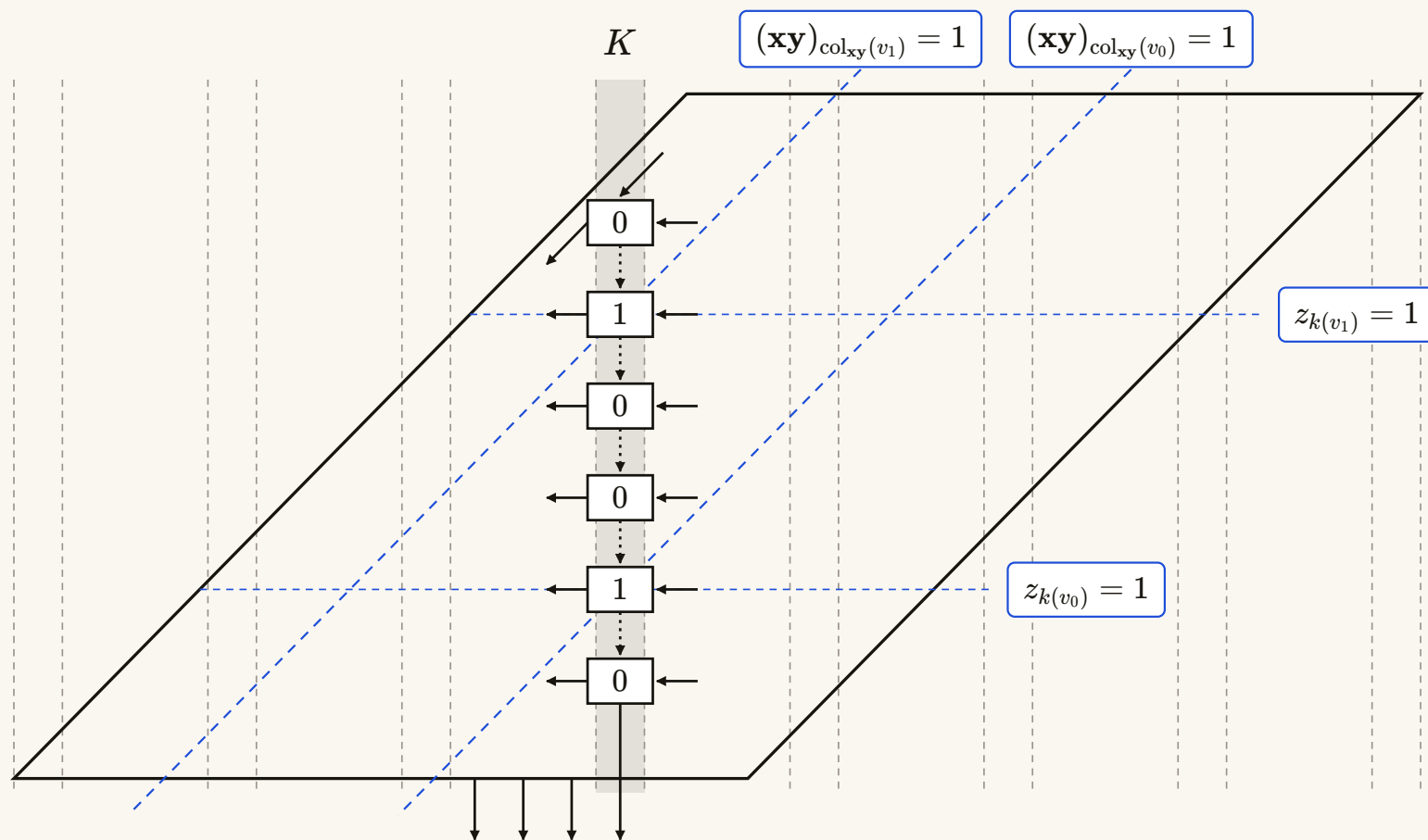
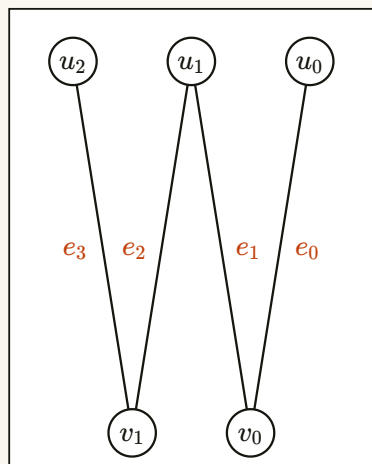
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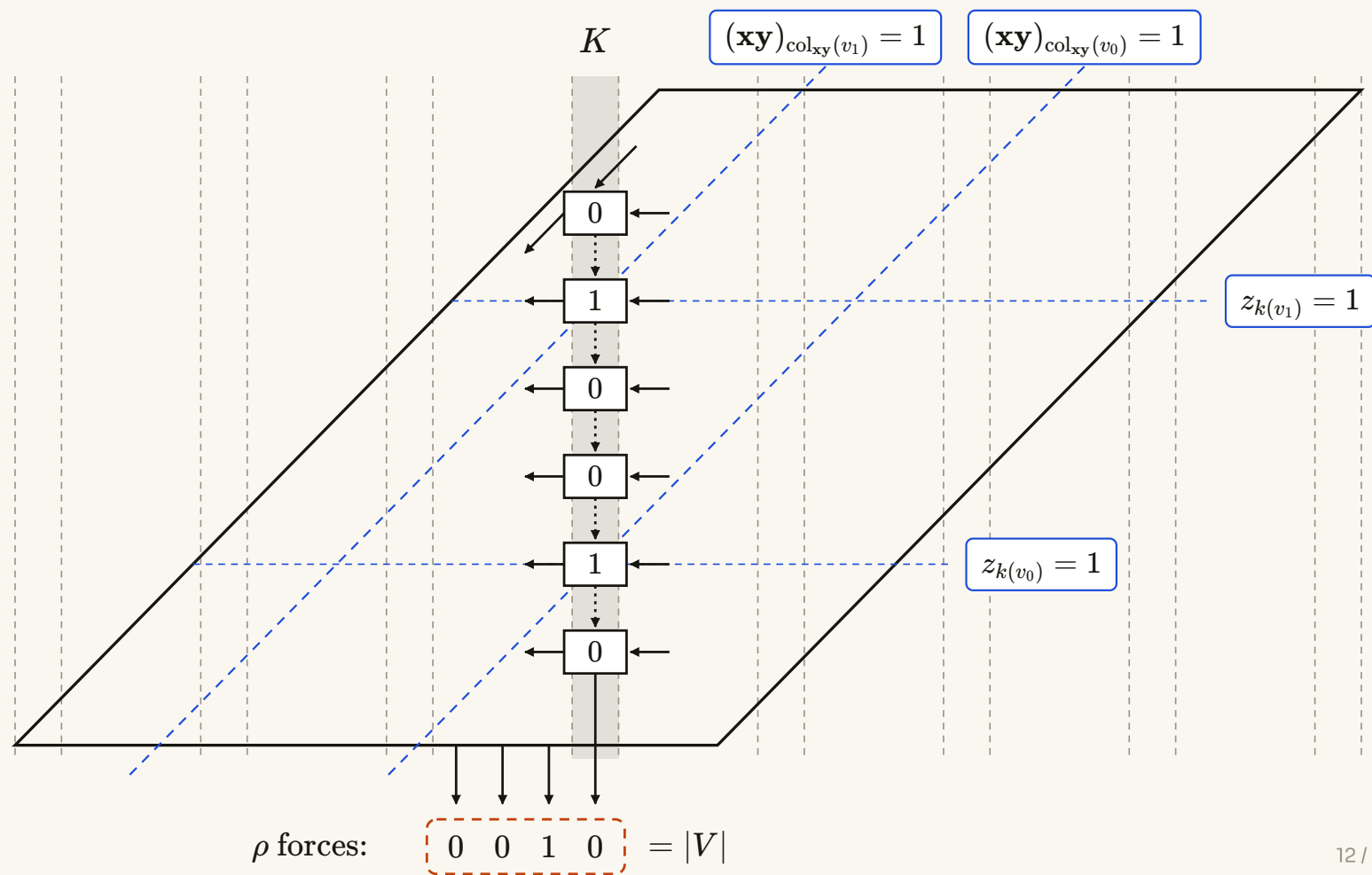
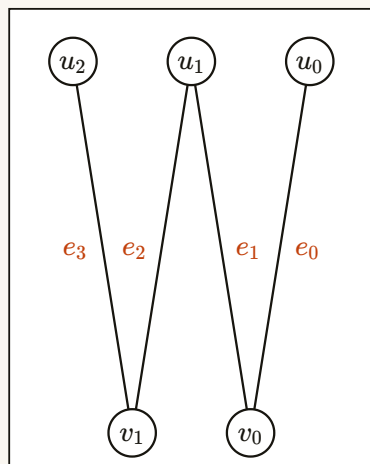
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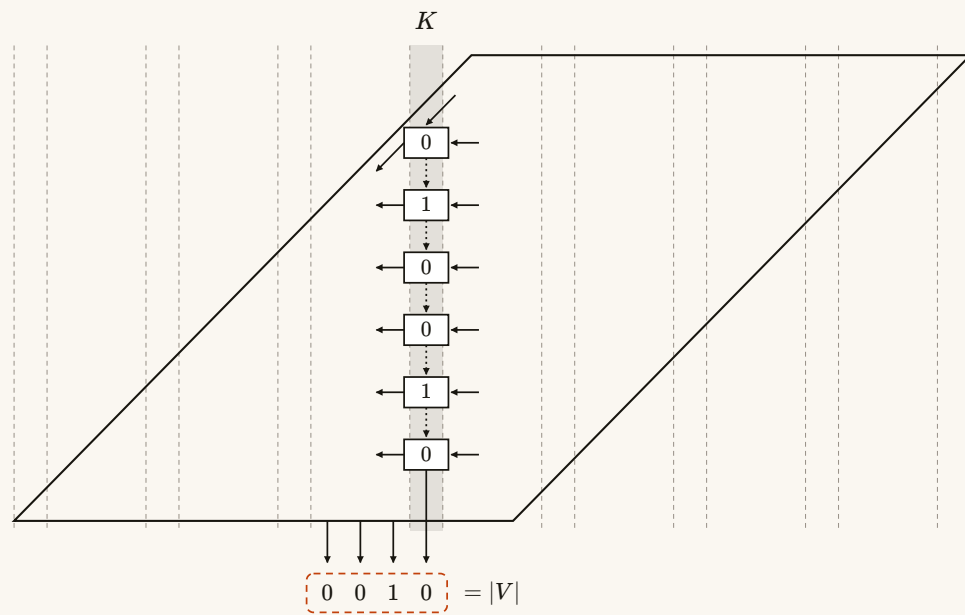


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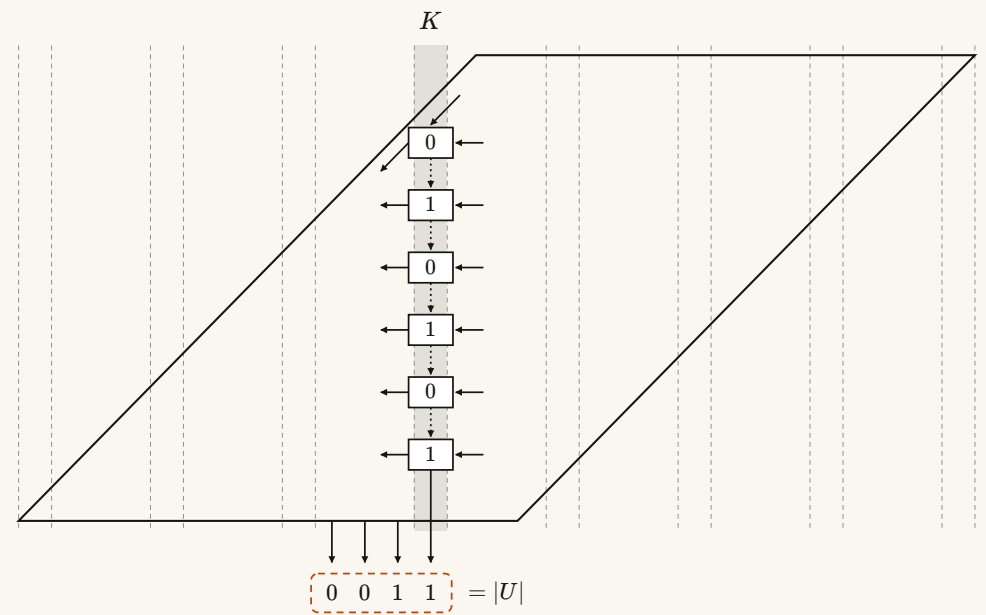


Restriction ρ in the outer multipliers

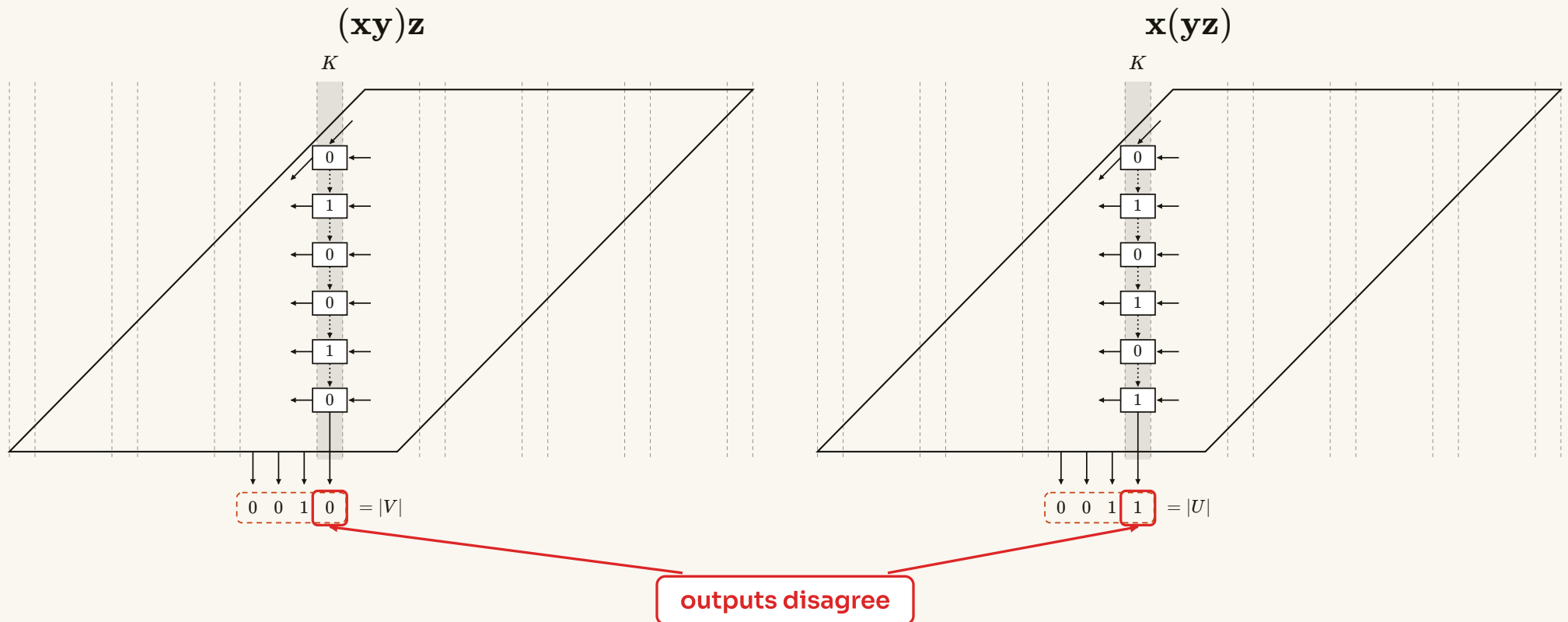
$(\mathbf{x}\mathbf{y})\mathbf{z}$



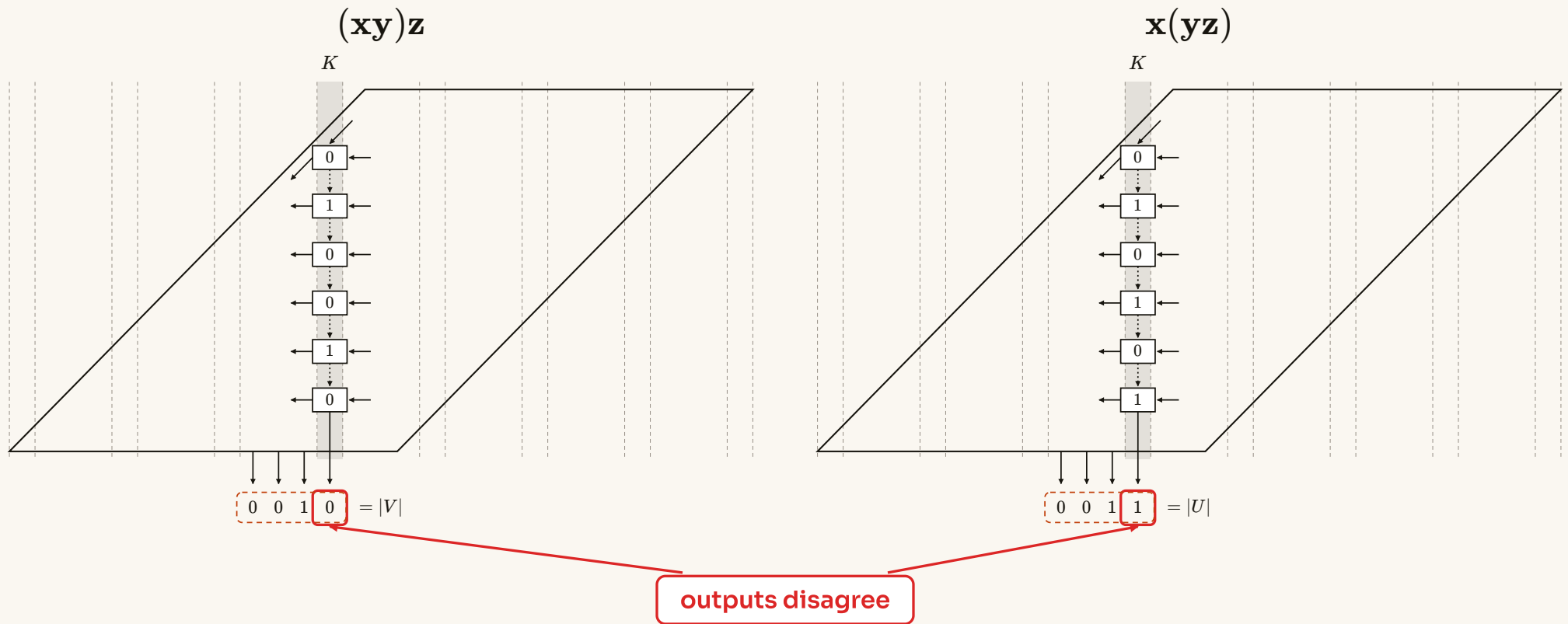
$\mathbf{x}(\mathbf{y}\mathbf{z})$



Restriction ρ in the outer multipliers



Restriction ρ in the outer multipliers



ρ sets $e_K = 1$, satisfying long miter clause $e_0 \vee e_1 \vee \dots \vee e_{4n-1}$

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SPM clauses:

$\text{ExactOne}(X_{E(w)}): \bigvee_{e \in E(w)} X_e$ and $\neg X_e \vee \neg X_f$ for $e, f \in E(w)$ where $e \neq f$

$\text{Def}(Z_{w,f}):$ canonical CNF for $Z_{w,f} \leftrightarrow f(X_{E(w)})$

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- C has constant width, so it depends on X_e from a constant number of stars $E(w)$
 - o i.e. $|\text{supp}(C)|$ is constant
- By implicational completeness, suffices to show α from lemma satisfies C^σ
 - o Indeed, for every X_e assignment, wire values substituted by σ agree with propagated values, i.e. $\alpha(\sigma(q)) = \bar{\alpha}(q)$ for variables $q \in C$

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Facts about ρ :

- ρ will restrict subset of bit-vectors $\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{xy}, \mathbf{yz}$, and set miter variable $e_K = 1$
- $\mathbf{xy} \upharpoonright \rho$ encodes, with auxiliary variables, constraints $\text{ExactOne}(X_{E(v)})$ for $v \in V$

Proof of local soundness lemma

Lemma 8.6 Let C be a clause of $\text{Assoc}_n \upharpoonright \rho$. If an assignment α to the X_e variables satisfies $\text{ExactOne}(X_{E(w)})$ for every $w \in \text{supp}(C)$, then the propagated assignment $\bar{\alpha}$ satisfies C .

Facts about ρ :

- ρ will restrict subset of bit-vectors $\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{xy}, \mathbf{yz}$, and set miter variable $e_K = 1$
- $\mathbf{xy} \upharpoonright \rho$ encodes, with auxiliary variables, constraints $\text{ExactOne}(X_{E(v)})$ for $v \in V$

Main Case: C came from inner multiplier $\mathbf{xy} \upharpoonright \rho$

Proof of local soundness lemma

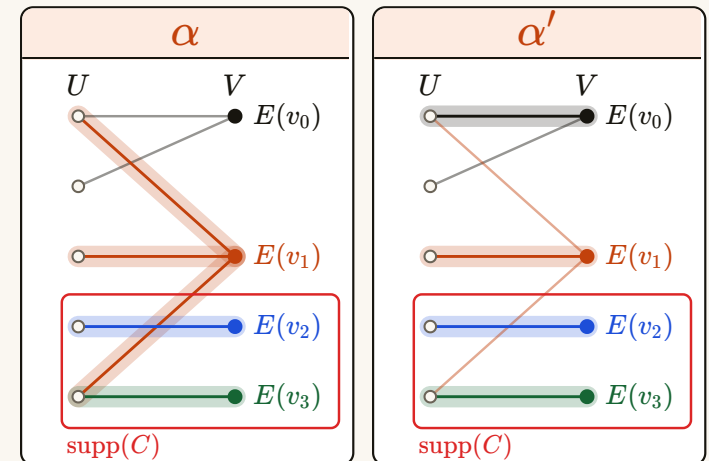
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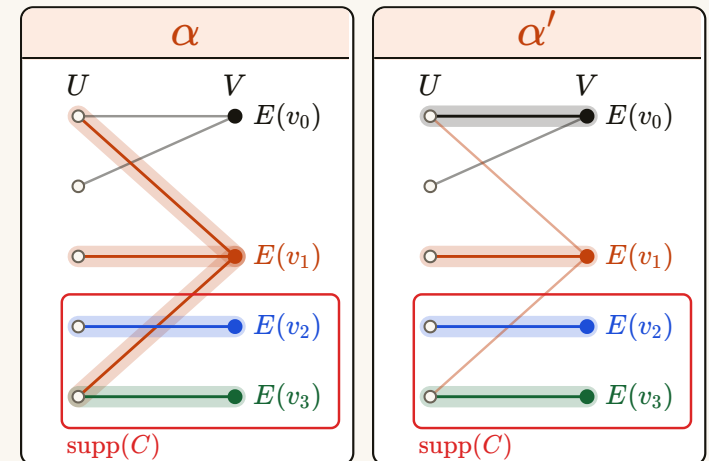
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Proof of local soundness lemma

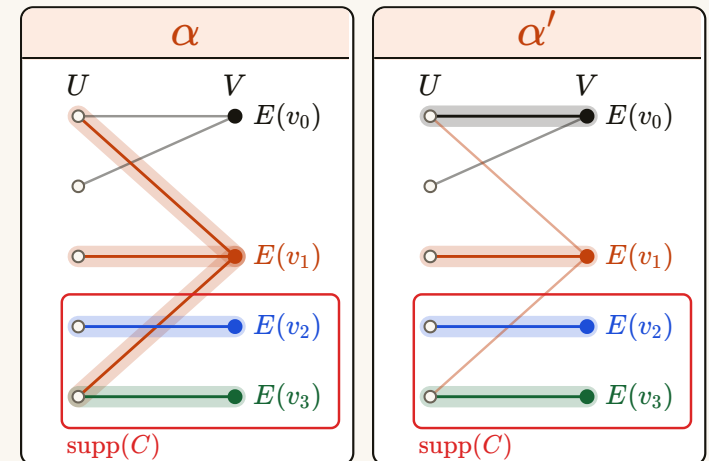
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- Let α' “repair” α by selecting one edge in $E(v)$ for $v \notin \text{supp}(C)$
- Propagated assignment $\bar{\alpha}'$ satisfies all clauses of $\mathbf{xy} \upharpoonright \rho$, including C
- $\bar{\alpha}'$ agrees with $\bar{\alpha}$ on C , by definition of $\text{supp}(C)$



Proof of local soundness lemma

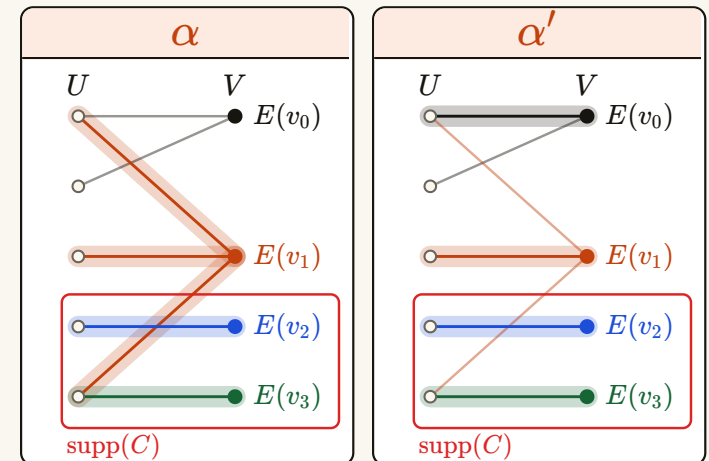
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- Let α' “repair” α by selecting one edge in $E(v)$ for $v \notin \text{supp}(C)$
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- $\bar{\alpha}'$ agrees with $\bar{\alpha}$ on C , by definition of $\text{supp}(C)$
- Hence $\bar{\alpha}$ also satisfies C



Proof of local soundness lemma

Lemma 8.6 Let C be a clause of $\text{Assoc}_n \upharpoonright \rho$. If an assignment α to the X_e variables satisfies $\text{ExactOne}(X_{E(w)})$ for every $w \in \text{supp}(C)$, then the propagated assignment $\bar{\alpha}$ satisfies C .

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Remaining Cases:

Proof of local soundness lemma

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Remaining Cases:

- C came from $\mathbf{yz} \upharpoonright \rho$ (symmetric proof as $\mathbf{xy} \upharpoonright \rho$ case)

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Remaining Cases:

- C came from $\mathbf{yz} \upharpoonright \rho$ (symmetric proof as $\mathbf{xy} \upharpoonright \rho$ case)
- C came from outer multiplier, e.g. $\mathbf{x}(\mathbf{yz}) \upharpoonright \rho$
 - o ρ only restricted inputs $\mathbf{x}$ and $\mathbf{yz}$. So propagating α satisfies all clauses of $\mathbf{x}(\mathbf{yz}) \upharpoonright \rho$

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Remaining Cases:

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- C came from outer multiplier, e.g. $\mathbf{x}(\mathbf{yz}) \upharpoonright \rho$
 - o ρ only restricted inputs $\mathbf{x}$ and $\mathbf{yz}$. So propagating α satisfies all clauses of $\mathbf{x}(\mathbf{yz}) \upharpoonright \rho$
- C was a clause defining a miter variable e_ℓ
 - o Miter clause gone, so no constraint on e_ℓ . So $\bar{\alpha}$ satisfies C

Construction of ρ : setup

Requirements for ρ

- Inner multipliers impose perfect matching constraints on X_e variables
- Outer multipliers disagree at fixed output bit K
- Every column contains X_e from at most one star

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Setup

Build index maps $i(u), j(e), k(v)$

- u, e, v index bits of $\mathbf{x}, \mathbf{y}, \mathbf{z}$

	8				4				0			
x:	1	0	0	0	1	0	0	0	1			
	↑				↑				↑			
	u_2				u_1				u_0			
y:	X_{e_2}	0	0	0	X_{e_1}	0	0	0	X_{e_0}			
	↑				↑				↑			
	e_2				e_1				e_0			
z:	1	0	0	0	0	0	0	0	1			
	↑								↑			
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Define $\text{col}_{\mathbf{xy}}(u, e) := i(u) + j(e)$ and $\text{col}_{\mathbf{yz}}(e, v) := j(e) + k(v)$

- Pair (u, e) indexes $\mathbf{xy}$ variable $t_{i(u), j(e)} = x_{i(u)} \wedge y_{j(e)}$
- Pair (e, v) indexes $\mathbf{yz}$ variable $t_{j(e), k(v)} = y_{j(e)} \wedge z_{k(v)}$

	8				4				0			
x:	1	0	0	0	1	0	0	0	1			
	↑				↑				↑			
	u_2				u_1				u_0			
y:	X_{e_2}	0	0	0	X_{e_1}	0	0	0	X_{e_0}			
	↑				↑				↑			
	e_2				e_1				e_0			
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Define $\text{col}_{(\mathbf{xy})\mathbf{z}}(u, e, v) := \text{col}_{\mathbf{xy}}(u, e) + k(v) := i(u) + j(e) + k(v)$

- Triple (u, e, v) indexes $(\mathbf{xy})\mathbf{z}$ variable $t_{\text{col}_{\mathbf{xy}}(u, e), k(v)} = (\mathbf{xy})_{\text{col}_{\mathbf{xy}}(u, e)} \wedge z_{k(v)}$
- Triple (u, e, v) indexes $\mathbf{x}(\mathbf{yz})$ variable $t_{i(u), \text{col}_{\mathbf{yz}}(e, v)} = x_{i(u)} \wedge (\mathbf{yz})_{\text{col}_{\mathbf{yz}}(e, v)}$

	8				4				0			
x:	1	0	0	0	1	0	0	0	1			
	↑				↑				↑			
	u_2				u_1				u_0			
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	↑				↑				↑			
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$$\text{col}_{\mathbf{x}\mathbf{y}}(u, e) := i(u) + j(e) \quad \text{col}_{\mathbf{y}\mathbf{z}}(e, v) := j(e) + k(v)$$

$$\text{col}_{(\mathbf{x}\mathbf{y})\mathbf{z}}(u, e, v) := \text{col}_{\mathbf{x}(\mathbf{y}\mathbf{z})}(u, e, v) := i(u) + j(e) + k(v)$$

Goal (Lemma 7.3):

(i) $i(u), j(e), k(v)$ are injective and multiples of g

- Makes the restriction g -sparse

	8				4				0			
x:	1	0	0	0	1	0	0	0	1			
	↑				↑				↑			
	u_2				u_1				u_0			
y:	X_{e_2}	0	0	0	X_{e_1}	0	0	0	X_{e_0}			
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Construction of ρ : Lemma 7.3 implies requirements

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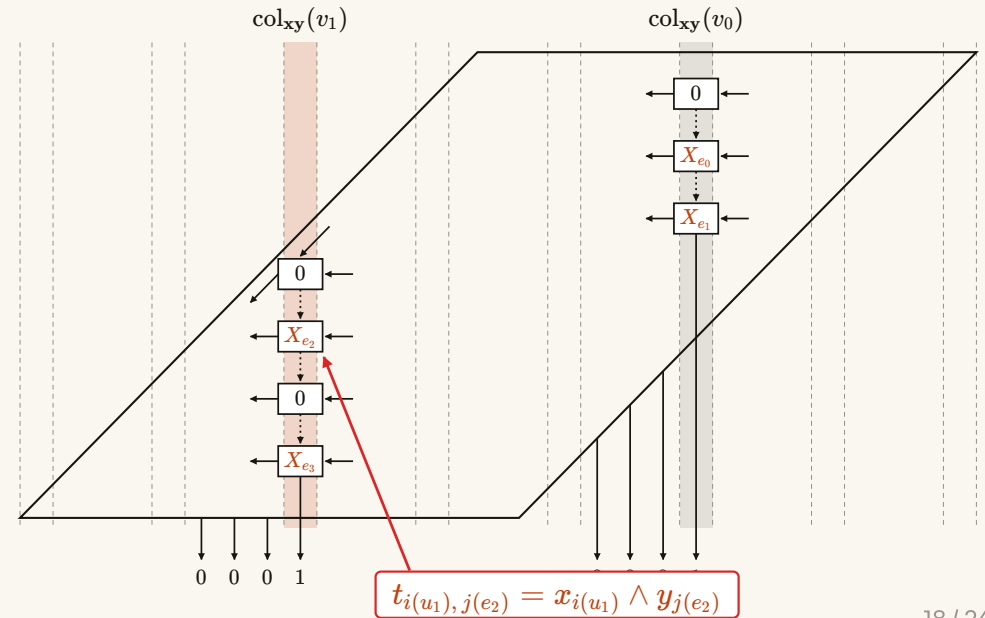
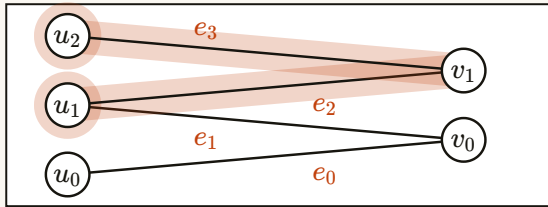
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Goal (Lemma 7.3):

(ii) Inner collision pattern:

- For $v \in V$, col_{xy} maps incident pairs (u_e, e) , $e \in E(v)$, to the same column, denoted $\text{col}_{xy}(v)$
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Construction of ρ : Lemma 7.3 implies requirements

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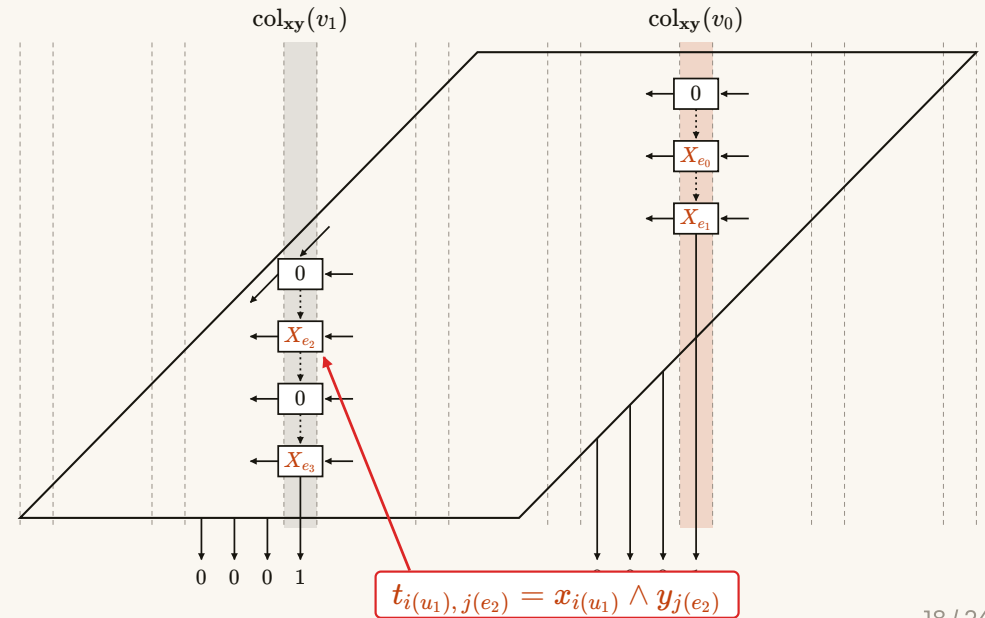
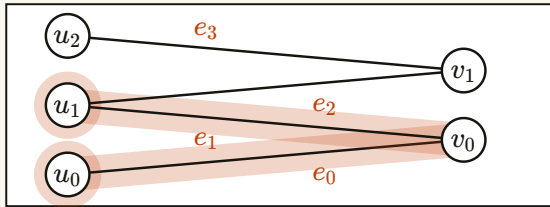
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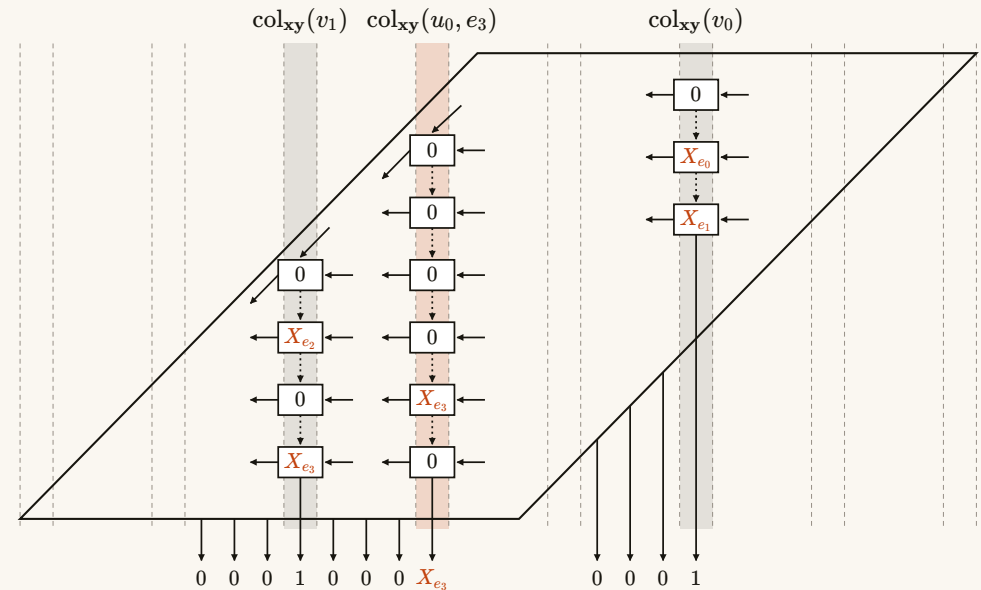
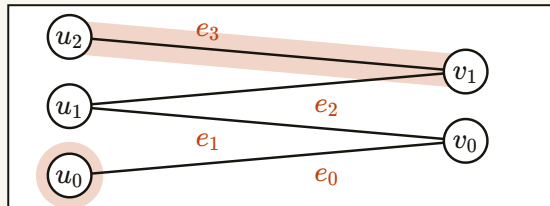
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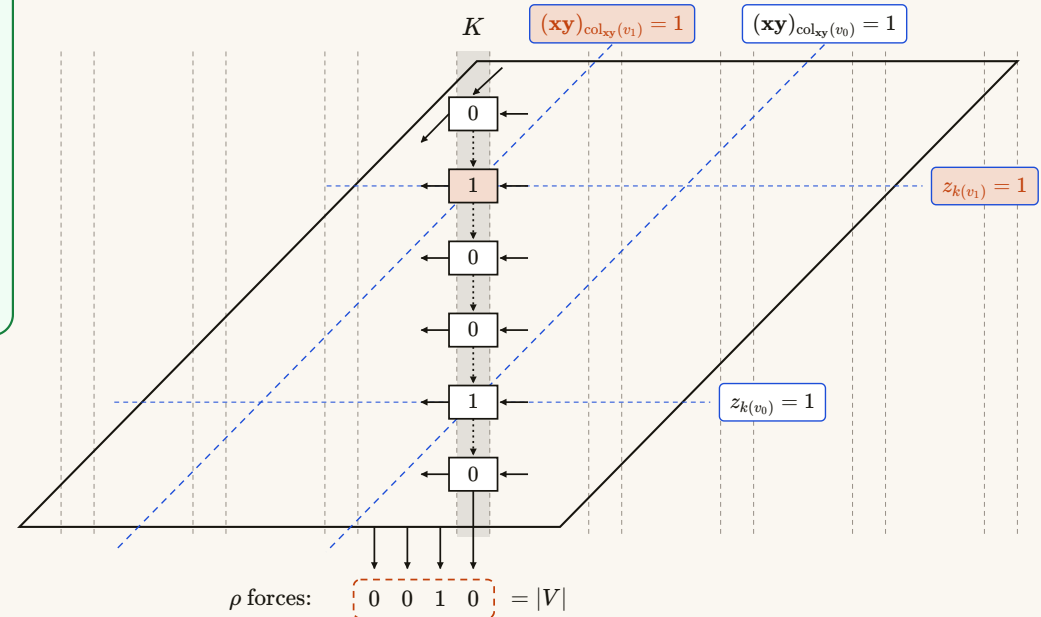
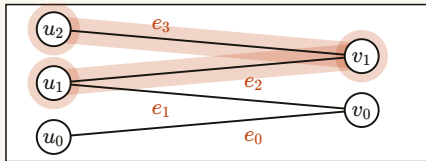
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- $\text{col}_{(xy)z}(u, e, v)$ is an injective fn of the difference pair
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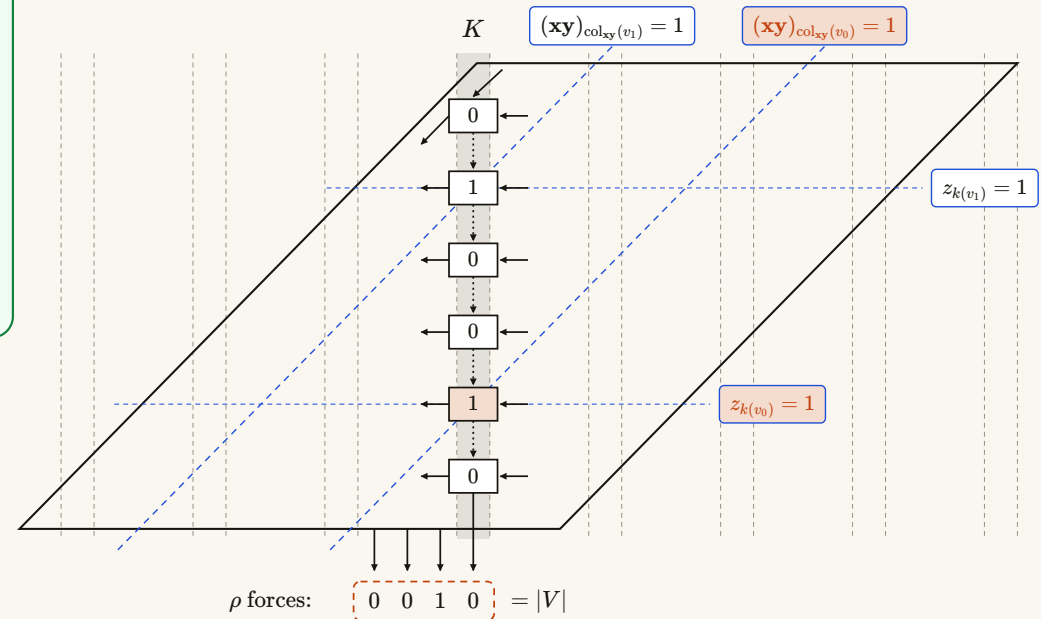
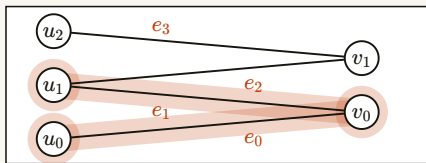
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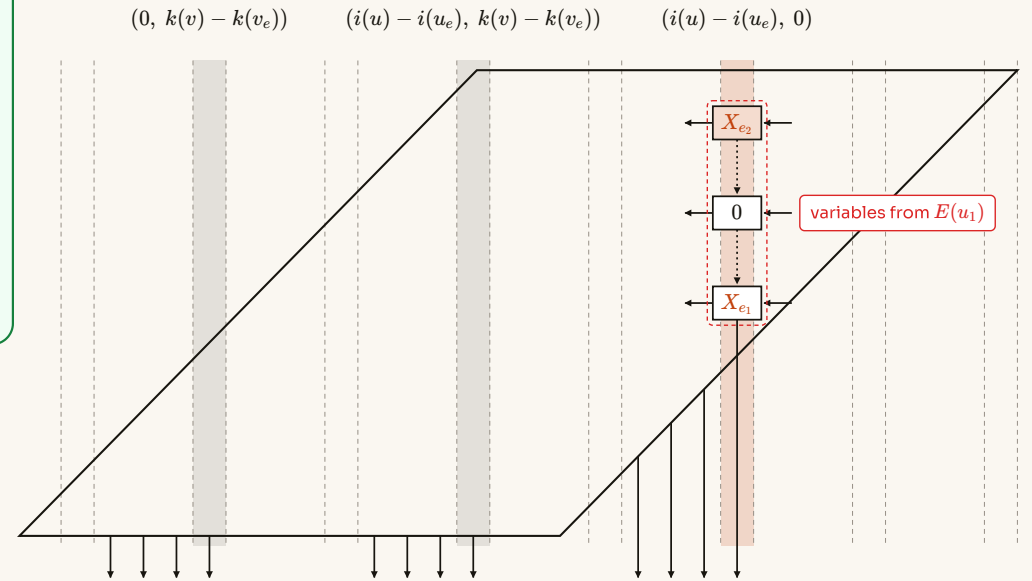
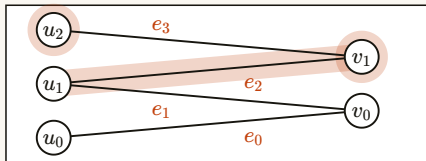
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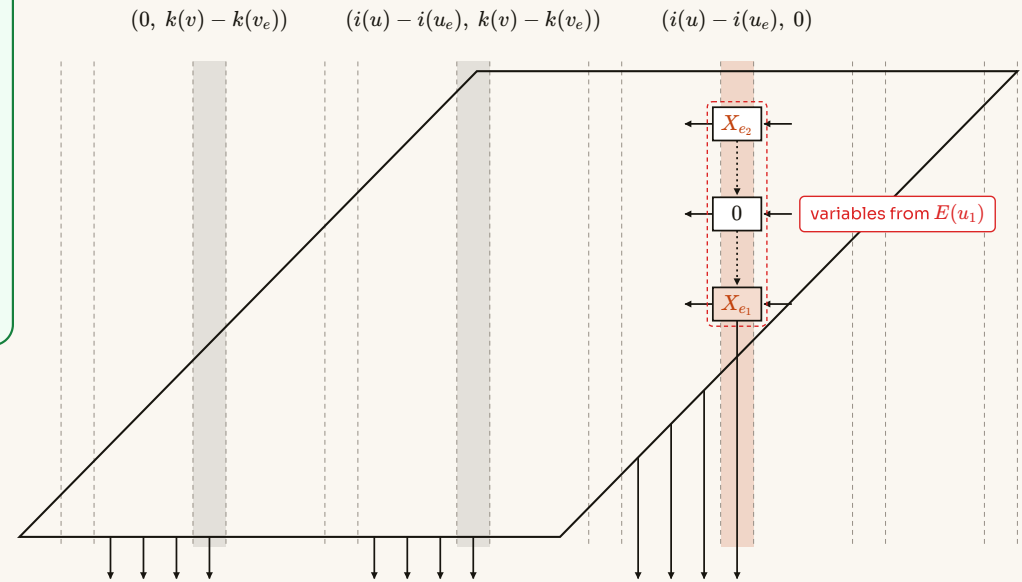
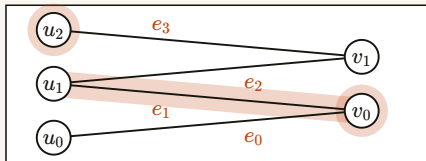
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Construction of ρ : Lemma 7.3 implies requirements

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- Inner multipliers impose perfect matching constraints on X_e variables
- Outer multipliers disagree at fixed output bit K
- Every column contains X_e from at most one star

$$\text{col}_{xy}(u, e) := i(u) + j(e) \quad \text{col}_{yz}(e, v) := j(e) + k(v)$$

$$\text{col}_{(xy)z}(u, e, v) := \text{col}_{x(yz)}(u, e, v) := i(u) + j(e) + k(v)$$

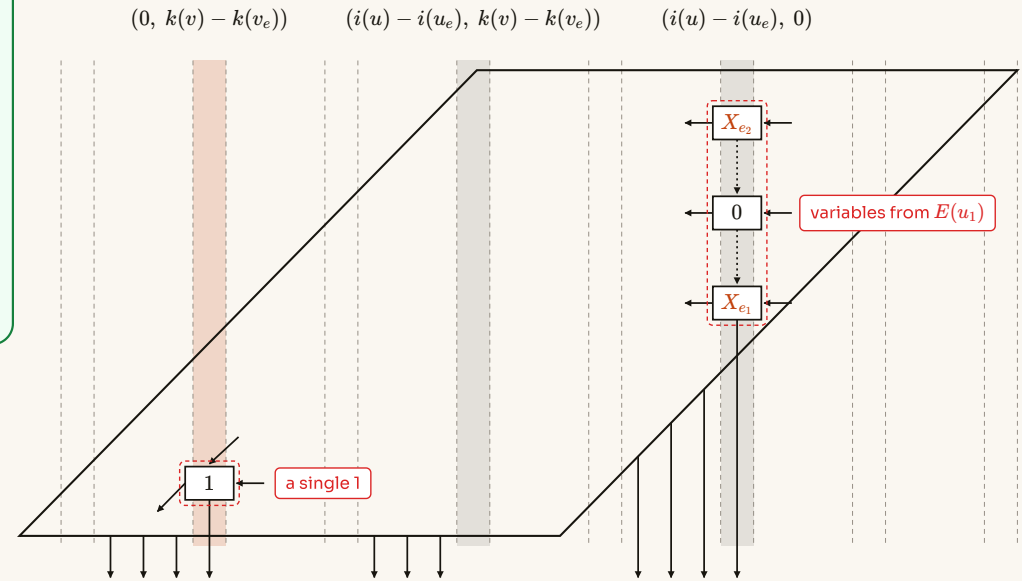
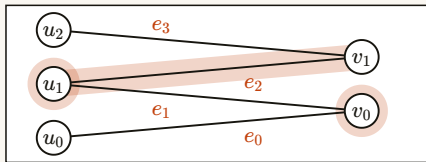
Goal (Lemma 7.3):

(iii) Outer collision pattern. Define difference pair of (u, e, v) :

$$(i(u) - i(u_e), k(v) - k(v_e))$$

- $\text{col}_{(xy)z}(u, e, v)$ is an injective fn of the difference pair
- Pair $(0, 0)$: Incident triples $(u_e, e, v_e) \mapsto$ fixed column K
- Pair $(i(u) - i(u_e) \neq 0, 0)$ comes from unique pair (u, u_e)
- Similar for $(0, k(v) - k(v_e))$ and $(i(u) - i(u_e), k(v) - k(v_e))$

- Triple $(u_e, e, v) \mapsto (\mathbf{xy})_{\text{col}_{xy}(v_e)} \wedge z_{k(v)} = 1$



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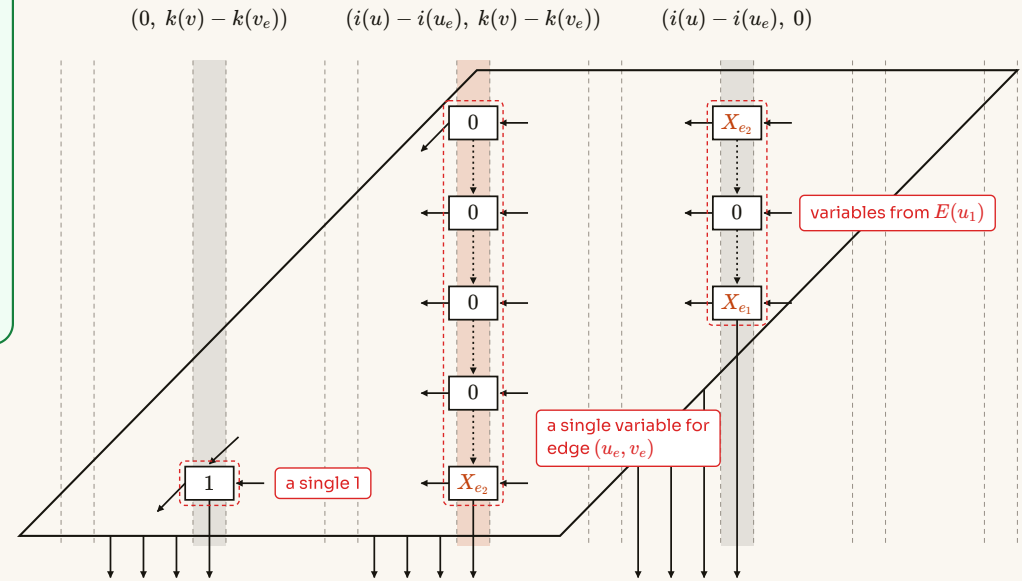
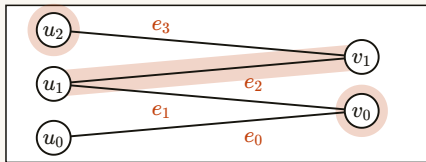
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- (i) $i(u), j(e), k(v)$ are injective and multiples of g
- (ii) **Inner collision pattern** (symmetric for $u \in U$ in yz):
 - For $v \in V$, col_{xy} maps incident pairs $(u_e, e), e \in E(v)$, to the same column
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Golomb ruler [Erdős–Turán 1941]

- Let $M = 32m^2$.
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- Let $M = 32m^2$.
- There are $m + 1$ integers $s_0, \dots, s_m \in [0, M)$, called *marks*, s.t. every nonzero difference d is produced by unique pair:

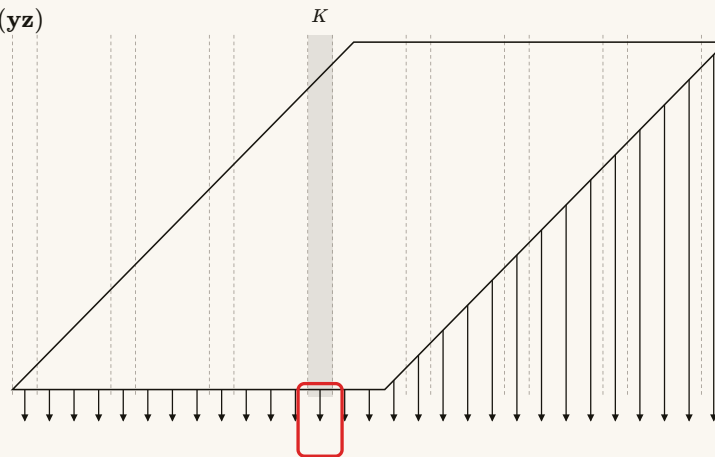
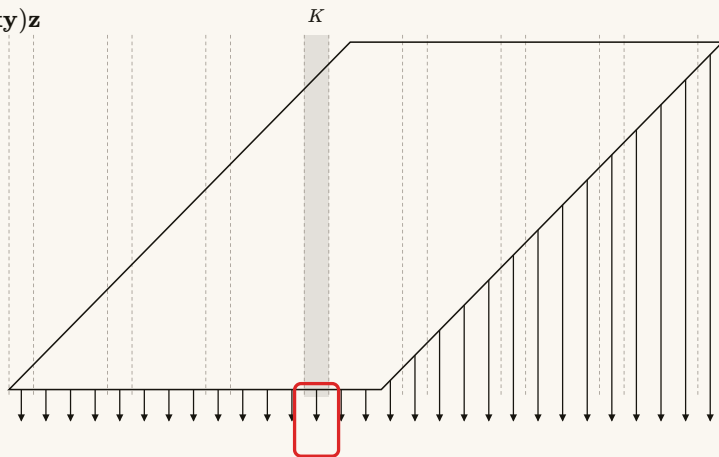
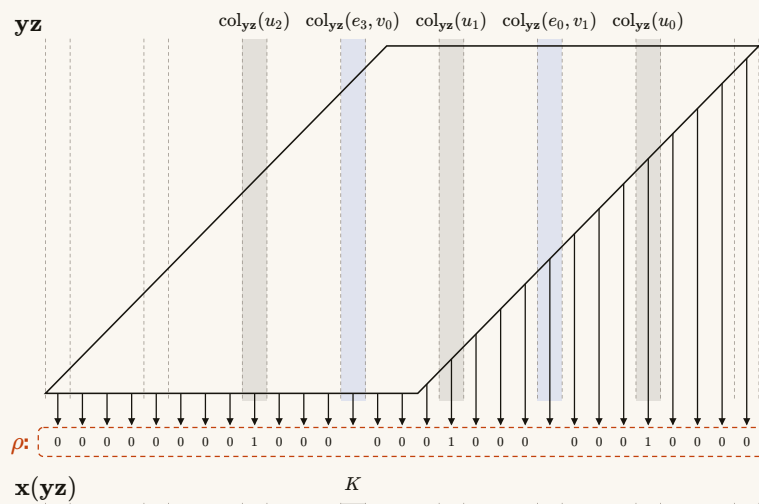
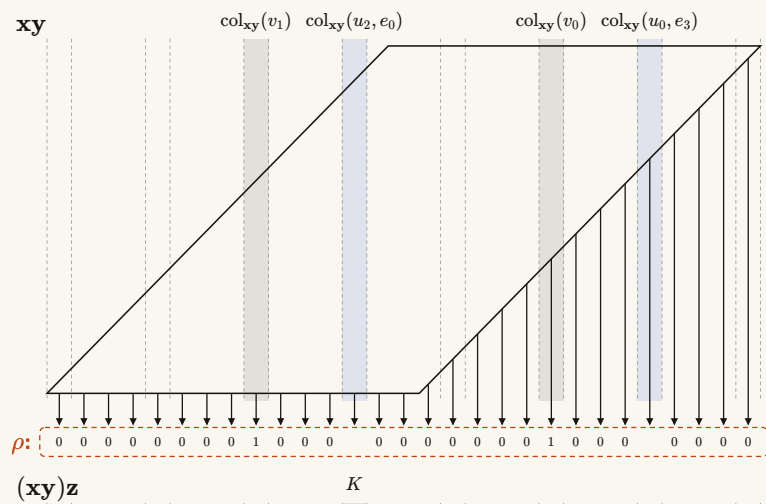
$$d = s_a - s_b$$

- Choose injection of U, V into the marks $s_0, \dots, s_m$. Write s_u, s_v for marks assigned to u and v

$$i(u) = gs_u, \quad k(v) = gBs_v, \quad j(e) = K - i(u_e) - k(v_e)$$

- $\text{col}_{xy}(u, e) = i(u) + j(e) = K + \underbrace{g(s_u - s_{u_e})}_{\Delta} - gBs_{v_e}$
 - Incident pair $(u_e, e) \Rightarrow \Delta = 0 \Rightarrow$ same column $K - gBs_{v_e}$
 - Non-incident pair $(u, e) \Rightarrow$ unique $\Delta \neq 0$ by Golomb ruler, unique s_{v_e} for edge $e \Rightarrow$ unique column

Construction of ρ



ρ sets $e_K = 1$

Recap of proof

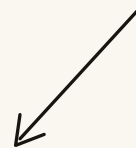
Lemma 7.3:

- (i) i, j, k injective, multiples of g
- (ii) Inner collision pattern
- (iii) Outer collision pattern



Requirements for ρ

- Inner multipliers impose perfect matching constraints on X_e variables
- Outer multipliers disagree at fixed output bit K
- Every column contains X_e from at most one star



Lemma 8.6 (local soundness). If α satisfies $\text{ExactOne}(X_{E(w)})$ for every $w \in \text{supp}(C)$, then $\bar{\alpha}$ satisfies C .



Constant size derivation of each $(\text{Assoc}_n \upharpoonright \rho)^\sigma$ clause from SPM clauses

